==========

PlayList : spring boot full course

channel : india free internship

youtube channel https://www.youtube.com/@IndiaFreeInternship

source Code: spring boot source Code

notes Link : https://github.com/indiafreeinternship/NOTES/tree/main/spring%20boot%20notes

live notes and live coding

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SESSION 1

---

1. SPRING FRAMEWORK
2. SPRING BOOT
3. SRPING DATA JPA
4. SPRING WEB MVC
5. SPRING REST
6. SPRING SECURITY
7. MICROSERVICE
8. SPRING BATCH
9. SPRING SCHEDULAR

==========

SESSION 2

==========

Technolgies : webPage(servlets) +Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:

- Start coding from line zero
- Development takes more time
- More chance of error
- Design pattern applied manually

Framework:

- Ready-made structure
- In-built technologies
- Design patterns already implemented

RAD => Rapid Application Development

- Fast coding
- Less Error

Spring Framework:

- used for Web Application or Distributed appliation
- It has own Apis
- It Follow Container System

Spring Container

-Find/Scan Classes[spring bean]

- Create Object
- provide DATA

-Link Object based on Relation

- Finally Destroy

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	*programming*
oops	*core java*
ORM	*HIbernate with JPA*
DI	*Spring Ioc(Inversion of control)*

session 3

=========

Application

==========

3 Type of application

(i) Web Application

(ii) Distributed Application

(iii) Microservices Application

(i) Web Application : Collection of web pages run under webserver.

Application : - n services

Project :- n Module

eg : Gmail Application

User(Register and Login) , Inbox, Sent, Drfats, Setting, ... etc

Amazon Application

User(Register and Login) , search, Cart, Payment, ...etc

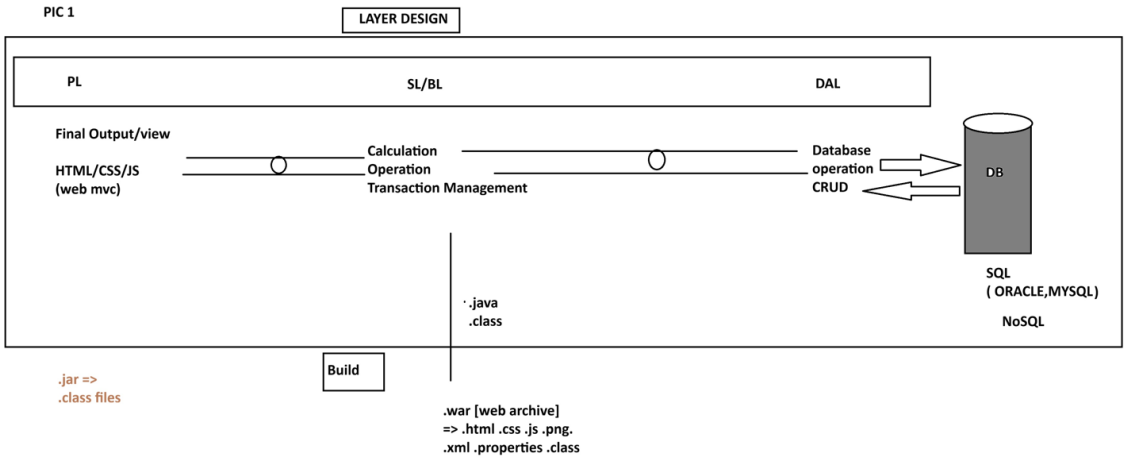
=> Implementation of service / module: - This is implemented using 3 Layers Design.

1. DAL : Data Access Layer (Database Operations)

2. SL : Service Layer (Calculation/Operations/txn mngmt)

3. PL : Presentation layer (view /Dynamic web pages/MVC)

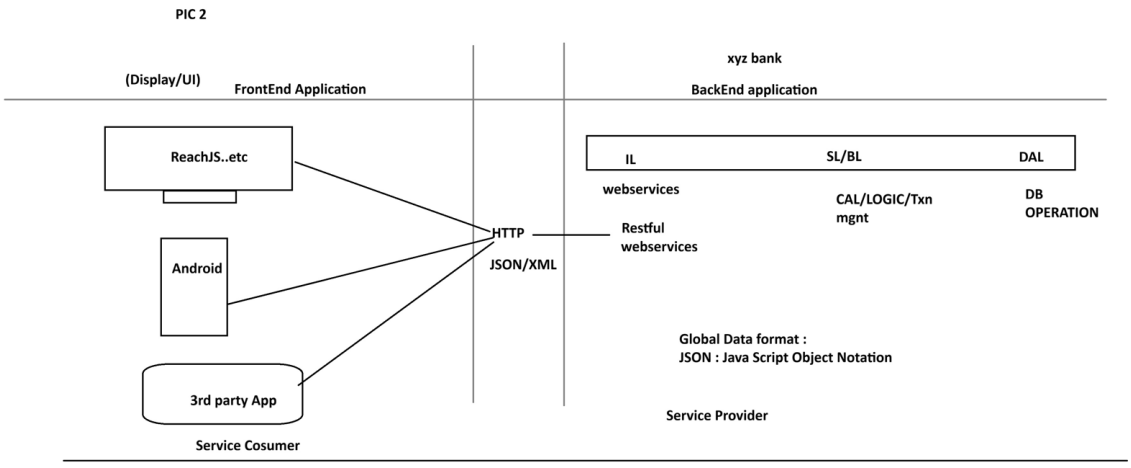
PIC 1



==========

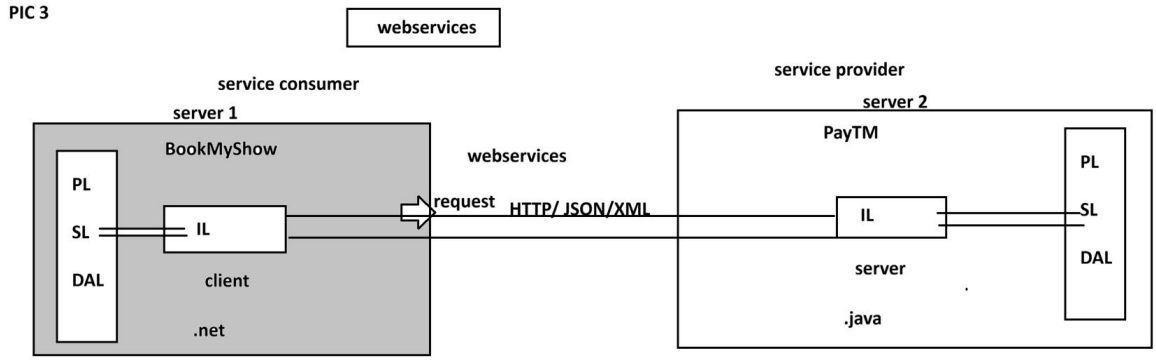
(ii) Distributed Application : An Application that runs at mutiple devices is called as distributed appliation.
xyz Bank application
This appliation can be accessed in different ways. Example.
eg:
1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)
.....

-But here business logics exist at backend application.
- Fontend applications are implemented using Angular, Android...etc
- App connected using HTTP Protocol using Global Data Format (JSON/XML)
PIC 2



WebServices:
==========
Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3



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session 4

==========
==========
SPRING FRAMEWORK
==========
CORE
---
1. DI and IOC
2. Container Types
3. Annotation Configuration
4. @ComponentScan
5. @Component
6. @Value
7. Java Configuration
8. @Configuration
9. @Bean
10. @PropertySource
11. Environment
12. Lifecycle methods
13. Runners
14. @ConfigurationProperties
15. Properties file
16. YAML File
17. Profiles
18. Lombok API

==========
spring core chapter 1
==========
Core:
This module/chapter gives all rules and guidelines to work with spring container.
1. DI and IOC
==========

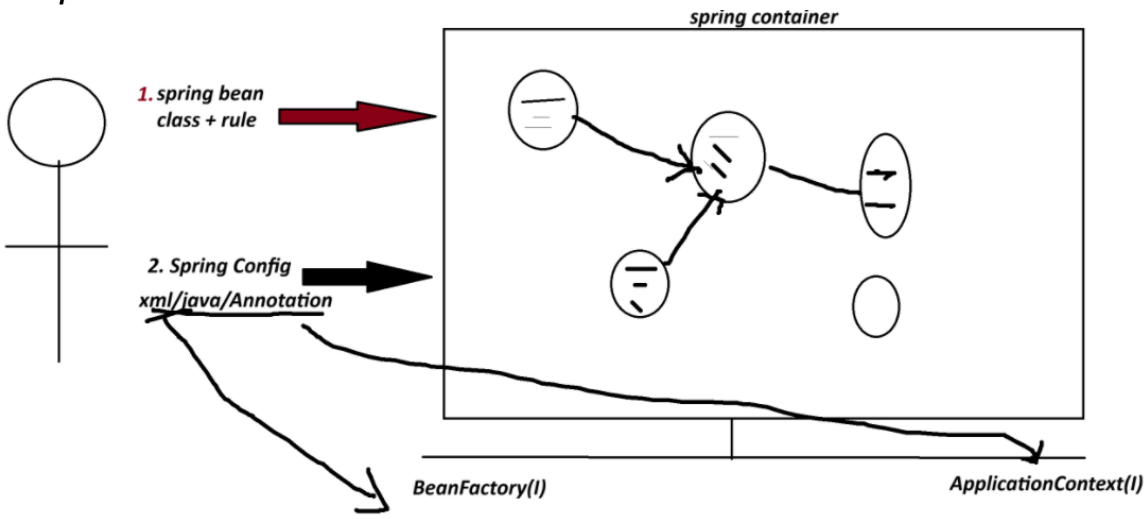
Spring Container :- It takes care of object.
1. Find/Scan our Classes(spring bean)
2. Create Object
3. Provide Data
4. Link Object
5. Destroy the Object

=>For this programmer has to provide two inputs-
1. Spring Bean : class+rules given by spring container.
2. Spring Configuration : This is main input cotains details like object name, relation with objects, data....etc.
xml/java/Annotation

*** Spring Container 2 Types:*
1. BeanFactory (Old container) : support only XML coonfiguration file.

2. ApplicationContext (New container) : support xml/java/Annotation configuration.

pic



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Dependeny Injection (DI)

Theory	*programming*
OOPs	*java*
ORM	*HIbernate with JPA*
DI	*Spring IoC(Spring Container) -> Inversion of Control*

=> Depedency :- An instance variable (Field) exist inside a class (spring bean)

1. Primitive Type Dependeny (PTD) :- If a variable is created using (8+1) datatype.
(byte, short, int, long, float, double, boolean, char +string)
(Their wrapper classess too)

2. Collection Type Depedency (CTD) :- If a variable created using(4) types:
(java.util.package)
(List,Map,Set and Properties)

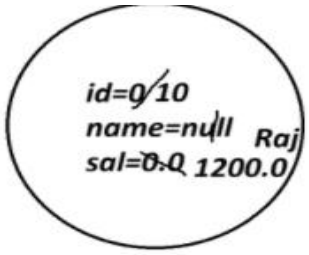
3. Reference Type Depedency (RTD) :- If a variable is creatd using other class/interface

Injection(4)
======
=> It give/provide/assign existed provide data to Depedency(Inside object)
1. Setter Injection (SI) :- provide data to variable using its set method(setter).

2. Constructor Injection(CI): provide data creating object using parameterized constructor.

3. LookUpMethod Injection(LMI) :- Spring bean Scope
[When parent bean is sigleton and child bean is prototype)
4. Interface Injection(II) : - Not Exist in Spring F/W.

```
class Employee{
   int id;
   String name;
   double sal;
}
```



Employee e=new Employee()

```
class Product{
   int id; //PTD
   String name; //PTD
   Boolean exist; //PTD

   List<String> models; //CTD
   Map<String,Integer> data; //CTD

   Vendor vob; //RTD
   Service sob; //RTD
}
```

```
class Vendor{}

interface Service{}
```

SESSION 5

==========

1. Spring Bean

2. STS Installation

Spring Bean

=========

-It is class that follow rules given by Spring Container.

Rule

-----

a. class must be public Type.

b. Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

c. Fields(Variables) are option, if required recomended to keep private.

private Integer empId;

private String empName;

d. If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{
private Integer id;
private String name;
public Empoyee(){
}
 public void setId(Integer id){
this.id=id;
 }

public Integer getId(){
 return id;
 }
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

==========

step 1:

go to official website

spring.io

link : - https://spring.io/tools#eclipse

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

==========

session 6

=========

1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object details (name, data,links with other objects ... etc).

3. Test class=> Just find, container created or not? object is creatd or not?

==========

Spring core XML Tags

==========

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

----

1. To create Object

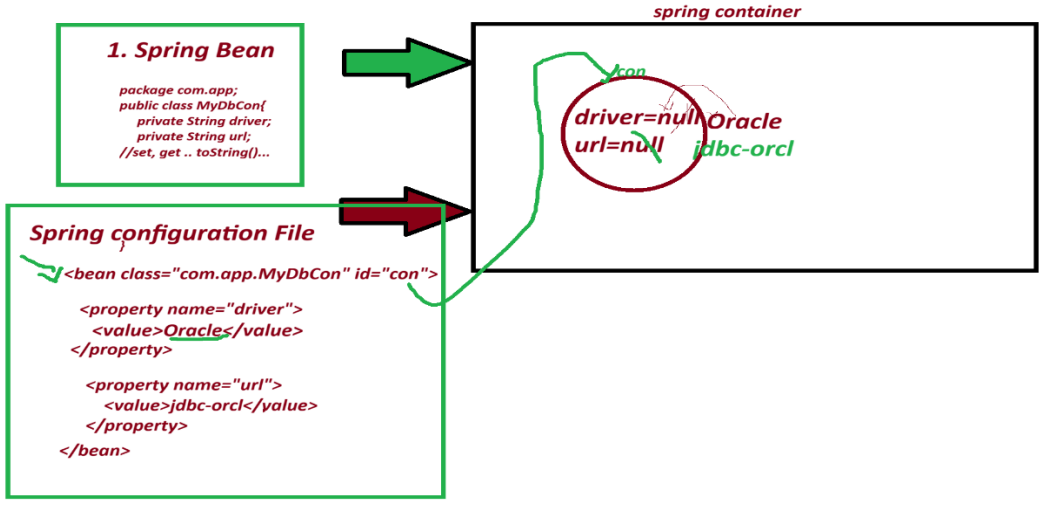
<bean id="objName" class="FullyQualifiedClassName"></bean>

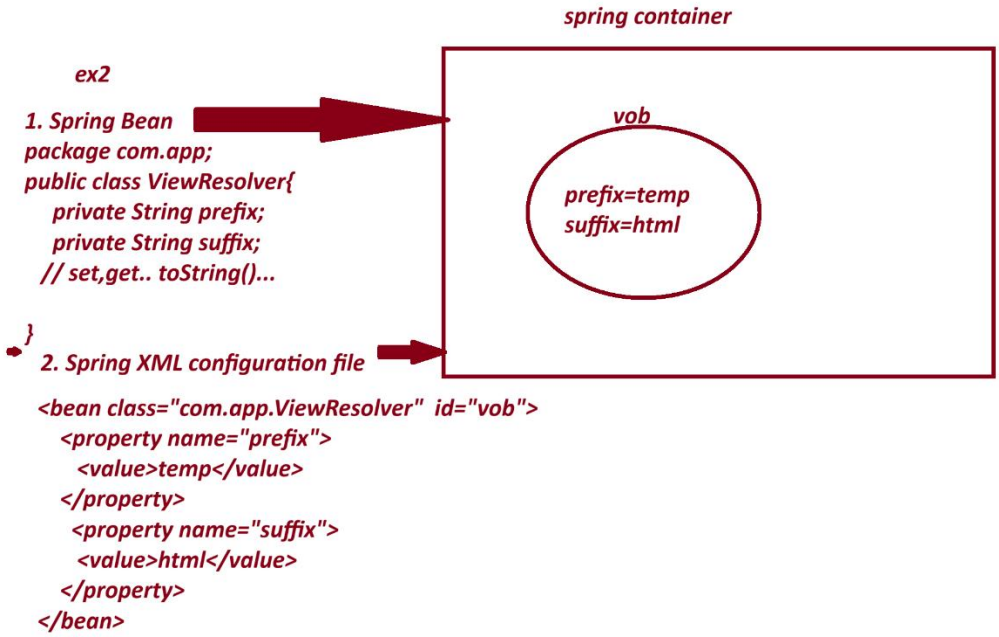
2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>





==========

1. Spring bean

```
package com.app;
public class PaymentService{
    private int token;
    private boolean active;
    private String provider;
   //set.. set toString..etc
}
```

2. Spring XML configuration file.

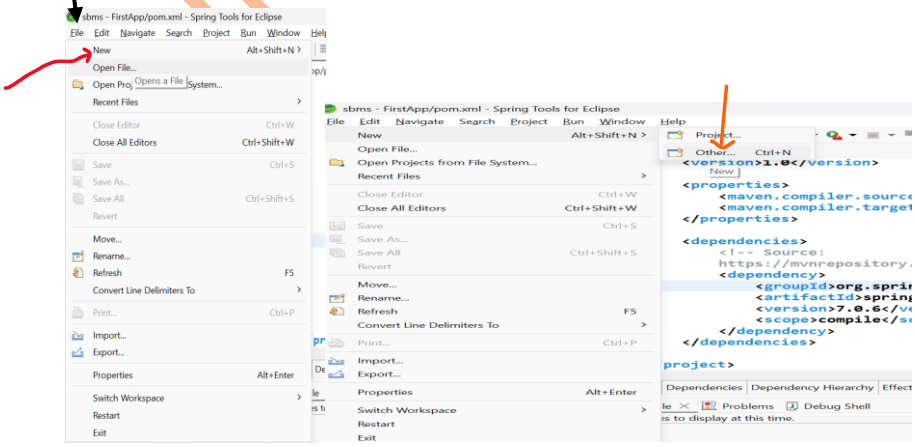
```
<bean class="com.app.PaymentService" id="psObj">
<property name="token">
<value>154678</value>
</property>
<property name="active">
<value>true</value>
</property>
<property name="provider">
<value>xyzBank</value>
</property>
</bean>
```

==========

CREATE MAVEN PROJECT WITH BELOW STEP :-

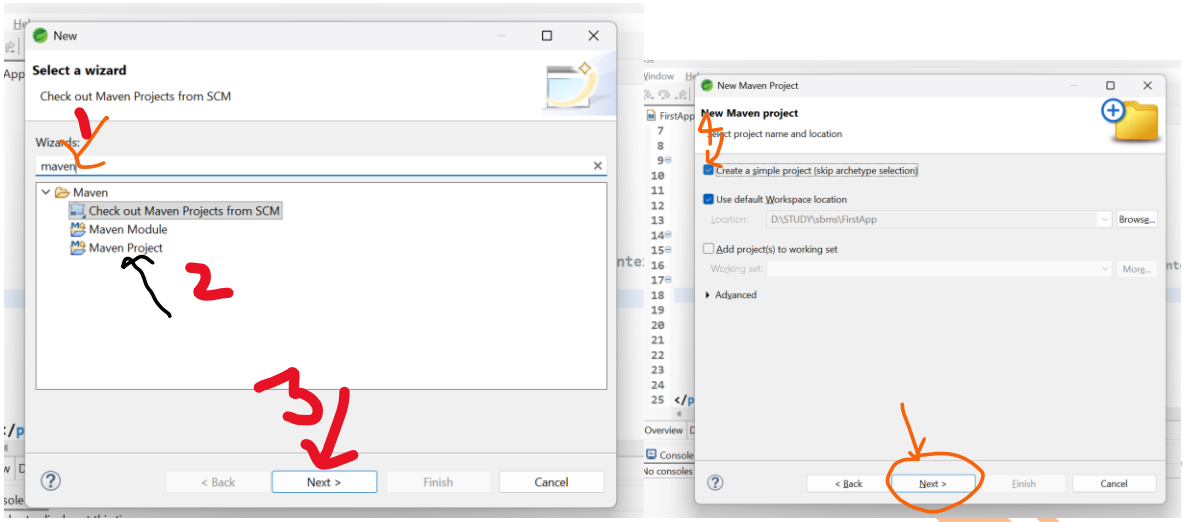
==========

1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

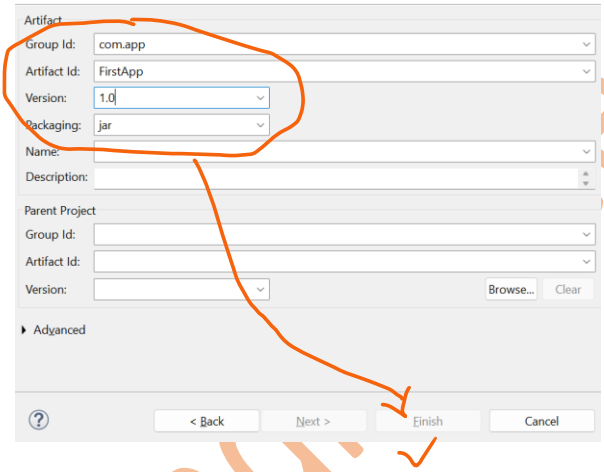
choose checkbox [v] create simple project -> next-> Enter details



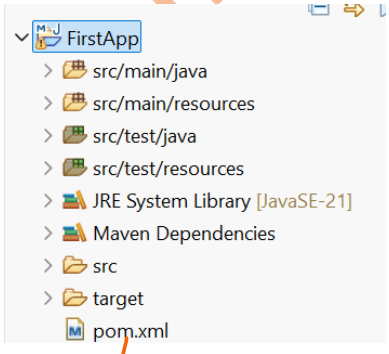
groupId : com.app

ArtficatId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>FirstApp</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
        <!-- Source:
         https://mvnrepository.com/artifact/org.springframework/spring-context -->
  <dependency>
    <groupId>org.springframework</groupId>
     <artifactId>spring-context</artifactId>
     <version>7.0.6</version>
     <scope>compile</scope>
  </dependency>
</dependencies>
</project>
```

---

session 7

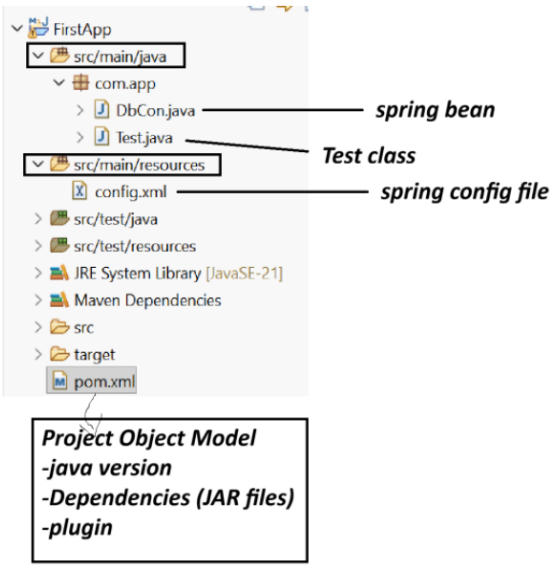
==========

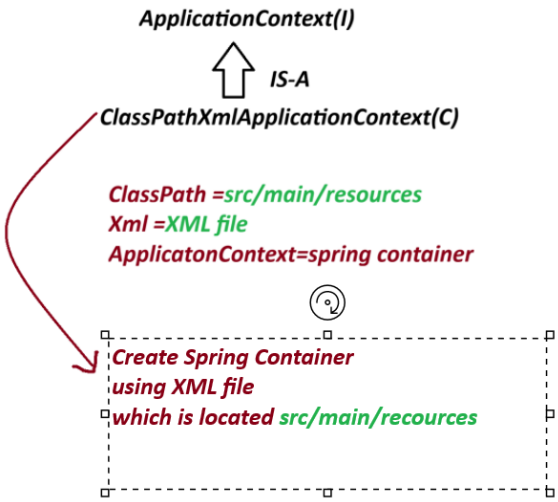
Spring Core First Application 3 Files.

1. Spring bean (.java)

2. Spring XML configuration (config.xml)

3. Test class(.java)





Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

==========

Create Maven Project with below Step:

==========

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtficatId - SecondApp

version - 1.0

----

code

SecondApp

==========

pom.xml

==========

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

DbCon.java

```
package com.app;

public class DbCon {
  private String driver;
  private String url;
```

```
  //alt+shit+s
  public DbCon() {
super();
System.out.println("Constructor called");
  }
  public String getDriver() {
return driver;
  }
  public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
  }
  public String getUrl() {
return url;
  }
  public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
  }
  @Override
  public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "]";
  }

}
==========================================

config.xml
----------
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

   <bean class="com.app.DbCon" id="mobj">
      <property name="driver">
      <value>Oracle Driver</value>
      </property>

      <property name="url">
      <value>JDBC-ORCL</value>
      </property>
   </bean>

</beans>
=======================

Test.java
----------------------
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
//ctrl+shit+O
public static void main(String[] args) {
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
Object ob = ac.getBean("mobj");
System.out.println(ob);
}
}
```

---

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

---

session 8

==========

Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

---

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

-

LookUpMethod Injetion

- Field Injection (Annotations)

==========

XML TAG

==========

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

==========

Collection Configuration

==========

=> Spring container support working wiht collection configuration which are from java.util package

Collection Name	Collection Type	XML Tag Name
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHasSet

Map -> LinkedHasMap

Properties --> NA

---

coding part collectionTypeS8

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>collectionTypeS8</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<dependencies><!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

MyDetails.java

```
package com.app;

import java.util.List;
import java.util.Map;
import java.util.Properties;
import java.util.Set;

public class MyDetails {
//ctrl+shift+O
private List<String> data;
private Set<String> models;
private Map<Integer,String> design;
private Properties myinfo;

public MyDetails() {
super();
```

```
}

public List<String> getData() {
return data;
}

public void setData(List<String> data) {
this.data = data;
}

public Set<String> getModels() {
return models;
}

public void setModels(Set<String> models) {
this.models = models;
}

public Map<Integer, String> getDesign() {
return design;
}

public void setDesign(Map<Integer, String> design) {
this.design = design;
}
public Properties getMyinfo() {
return myinfo;
}public void setMyinfo(Properties myinfo) {
this.myinfo = myinfo;
}
@Override
public String toString() {
```

```
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
"]";

}

}
```

---

xml config file

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.MyDetails" id="myinfo">

<property name="data">

<list>

<value>A</value>

<value>B</value>

<value>C</value>

<value>A</value>

</list>

</property>

<property name="models">

<set>

<value>M1</value>

<value>M2</value>

<value>M3</value>

<value>M1</value>

</set>

</property>
```

```
<property name="design">
<map>
<entry>
<key>
<value>101</value>
</key>
<value>AB</value>
</entry>
<entry>
<key>
<value>102</value>
</key>
<value>Suraj</value>
</entry>
<entry>
<key>
<value>103</value>
</key>

<null></null>
</entry>

</map>
</property>

<property name="myinfo">
<props>
<prop key="A1">B1</prop>
<prop key="A2">B2</prop>
<prop key="A3">B3</prop>
<prop key="A5"></prop>
</props>
```

```
</property>
</bean>
</beans>
```

Test File

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
//Object ob = ac.getBean("myinfo");
MyDetails ob = ac.getBean("myinfo",MyDetails.class);
System.out.println(ob);
System.out.println(ob.getData().getClass().getName());
System.out.println(ob.getModels().getClass().getName());
System.out.println(ob.getDesign().getClass().getName());
System.out.println(ob.getMyinfo().getClass().getName());
}

}
```

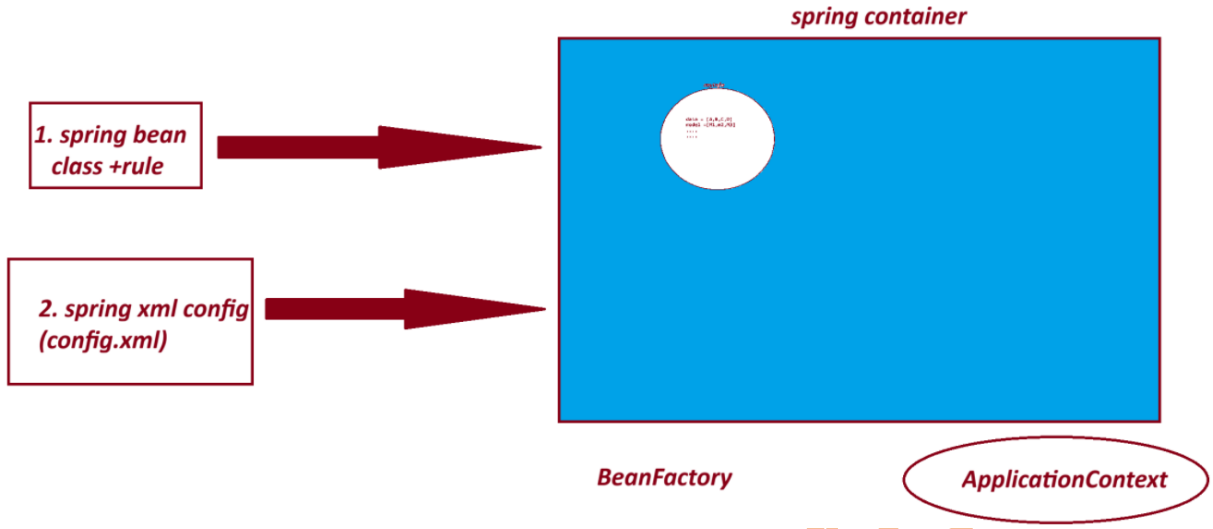
---

console output

```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null},
myinfo={A1=B1, A2=B2, A3=B3, A5=}]
java.util.ArrayList
java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

session 9

========

3. Reference Type Depedency - Setter Injection.

  ============================

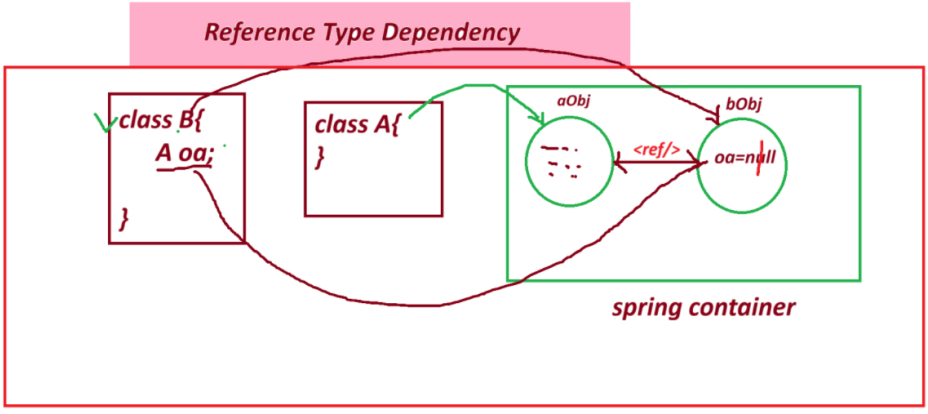
  => <ref/> : - tag is used to Link objects in case of HAS-A Relation.

        Syntax:

<property name="variableName">

<ref bean="childObjName"/>

</property/>



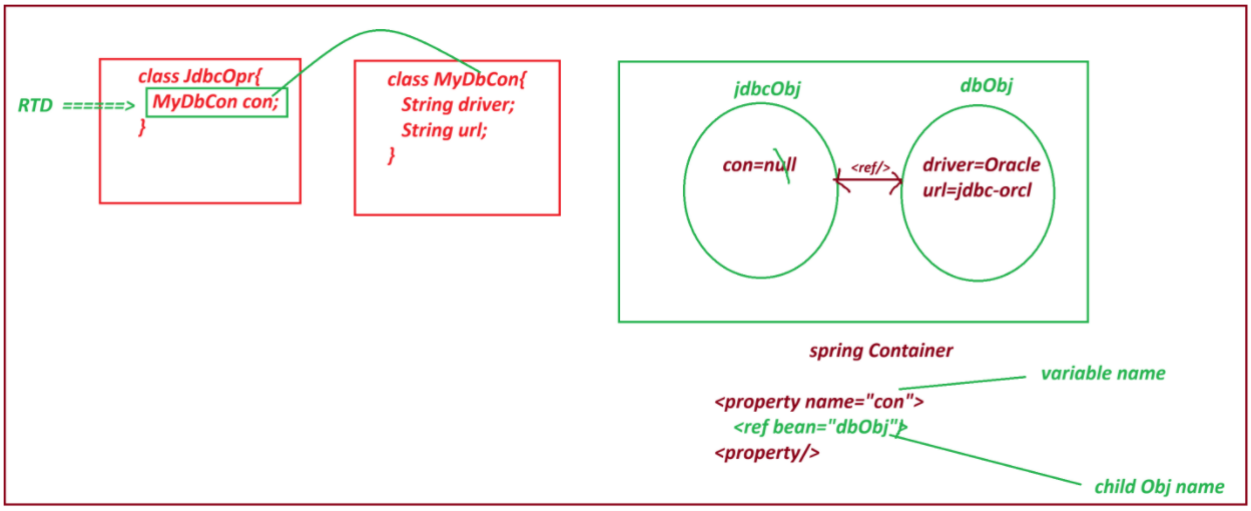
==========

- If we do not use <ref/> then ref variable holds default value null.

=> spring container creates objet to both classes and link them by using <ref/> tag.

=> if child object nam (ref bean name) is not valid or not exist then spring container throws exception like NoSuchBeanDefinitionException:

==========



project : ReferenceTypeS9

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ReferenceTypeS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

JdbcOpr.java

```
package com.app;

public class JdbcOpr {
  private MyDbCon con; //RTD

  public JdbcOpr() {
super();
// TODO Auto-generated constructor stub
  }

  public MyDbCon getCon() {
return con;
  }

  public void setCon(MyDbCon con) {
this.con = con;
  }
```

```
  @Override
  public String toString() {
return "JdbcOpr [con=" + con + "]";
  }

}
```

==========

MyDbCon.java

```
package com.app;

public class MyDbCon {
  private String driver;
  private String url;

  public MyDbCon() {
super();
// TODO Auto-generated constructor stub
  }

  public String getDriver() {
return driver;
  }

  public void setDriver(String driver) {
this.driver = driver;
  }

  public String getUrl() {
return url;
  }
```

```
 public void setUrl(String url) {
this.url = url;
 }

 @Override
 public String toString() {
return "MyDbCon [driver=" + driver + ", url=" + url + "]";
 }

 //alt+shift+s

}
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

 <bean class="com.app.MyDbCon"  id="dbObj" >
     <property name="driver">
     <value>OracleDriver</value>
     </property>
      <property name="url">
      <value>JDBC-ORCL</value>
      </property>
 </bean>
  <bean class="com.app.JdbcOpr"  id="jdbcObj">
    <property name="con">
    <ref bean="dbObj"/>
```

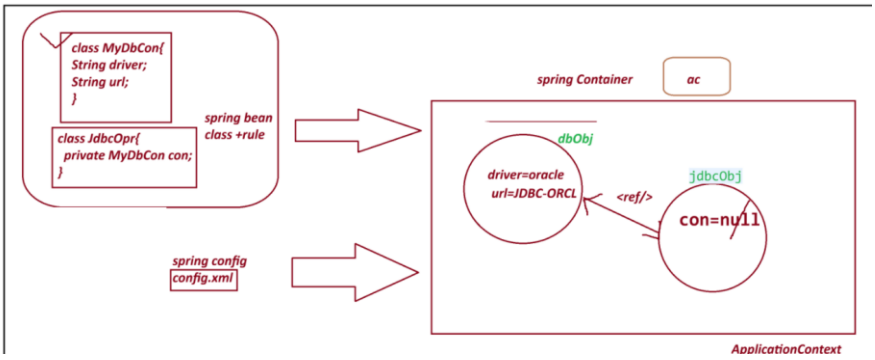
```
    </property>
  </bean>


</beans>
```

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
public static void main(String[] args) {
ApplicationContextac=new ClassPathXmlApplicationContext("config.xml");
JdbcOpr jdbc = ac.getBean("jdbcObj",JdbcOpr.class);
System.out.println(jdbc);
}
}
```

output

```
JdbcOpr [con=MyDbCon [driver=OracleDriver, url=JDBC-ORCL]]
==========================================
```



SI - Setter Injection :

Spring container creates object to given class using default constructor and provide data using setter method.

Employee emp=new Employee(); =======> <bean id="emp" class="Employee"/>

emp.setEmpId(10); =======> <property name="empId" value="10"/>

==========

***** Constructor Injection CI ************

==========

=> Sprint container creating object and providing data using Parameterized constructor.

Here , we use <constructor-arg> tag to indicate.

Employee emp=new Employee(10, "A"); =====>

```
<bean>
  <constructor-arg>
 <value>10</value>
</constructor-arg>
<constructor-arg>
 <value>A</value>
</constructor-arg>
</bean>
```

Note:

A. If a class has 3 param constructor then we need to pass <constructor-arg> 3 times in xml config file.

B. Data must be passed in order using <constructor-arg> as per constructor difined in class.

==========

code for CI (constructor Injection)

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ConstructorInjectionS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

Product.java

```
package com.app;

import java.util.List;

public class Product {

private String pcode;
```

```
private String model;
private List<String> data;
private Vendor vob;

public Product(String pcode, String model, List<String> data, Vendor vob) {
super();
this.pcode = pcode;
this.model = model;
this.data = data;
this.vob = vob;
}
@Override
public String toString() {
return "Product [pcode=" + pcode + ", model=" + model + ", data=" + data + ", vob=" + vob + "]";
}

}
```

Test.java

```
package com.app;

public class Vendor {

private String vcode;
private String vaddr;

public Vendor(String vcode, String vaddr) {
super();
this.vcode = vcode;
this.vaddr = vaddr;
}
```

```
@Override
public String toString() {
return "Vendor [vcode=" + vcode + ", vaddr=" + vaddr + "]";
}

}
```

---

config.xml

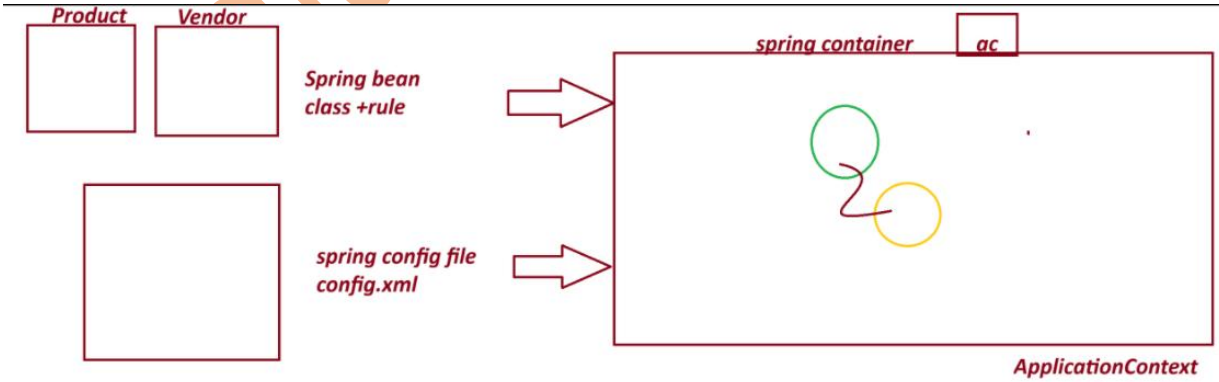
```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

 <bean class="com.app.Vendor" id="v1">
   <constructor-arg>
   <value>v1109</value>
   </constructor-arg>
   <constructor-arg>
   <value>IND</value>
   </constructor-arg>
 </bean>

 <bean class="com.app.Product" id="pobj">
  <constructor-arg>

<value>P01</value>
 </constructor-arg>


  <constructor-arg>
```

```
<value>HP-23444</value>
 </constructor-arg>

 <constructor-arg>
 <list>

<value>AA</value>

<value>BB</value>
 </list>
 </constructor-arg>

 <constructor-arg>
 <ref bean="v1"/>

</constructor-arg>
 </bean>

</beans>
```

output

=========

Product [pcode=P01, model=HP-23444, data=[AA, BB], vob=Vendor [vcode=v1109, vaddr=IND]]



==========

session 10

==========

Spring Annotation Configuration

This is Easy to Configure

Faster in execution compare to xml.

==========

1. Stereotype Annotation:

These annotation informs Spring Container to create Object.

They are 5 Annotation given by Spring Framework.

@Component : Create object inside container.

@Repostitory : Creates object + Database operations

@Service : Creates object + Transaction/logic/calculation..etc

@Controller : Creates object + Http Protocol (Web App)

@RestController : Creates object + Http Protocol (webservices)

@ComponentScan

@ComponentScans

2. Data Annotation/ Basic Annotation

@Value Annotation is used to provide data (field inject) to variable.

It has 3 usages:

1. Hard coding direct value to a variable.

2. Reading Data from Properties/YAML Files

3. - SpEL (Spring Expression language)

@Autowire

@Qualifier

@Primary

==========

ExcelExportAnnotationConfigS10

code

========

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ExcelExportAnnotationConfigS10</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean + annotation cofiguration

ExcelExport.java

```
package com.app;
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.stereotype.Component;
@Component("excel")
public class ExcelExport {
@Value("TEST")
   private String exportName;
@Value(".csv")
   private String fileExt;

@Override
public String toString() {
return "ExcelExport [exportName=" + exportName + ", fileExt=" + fileExt + "]";
}

}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.ComponentScans;

@ComponentScan("com.app")
public class MyAppConfig {

}
```

Test.java

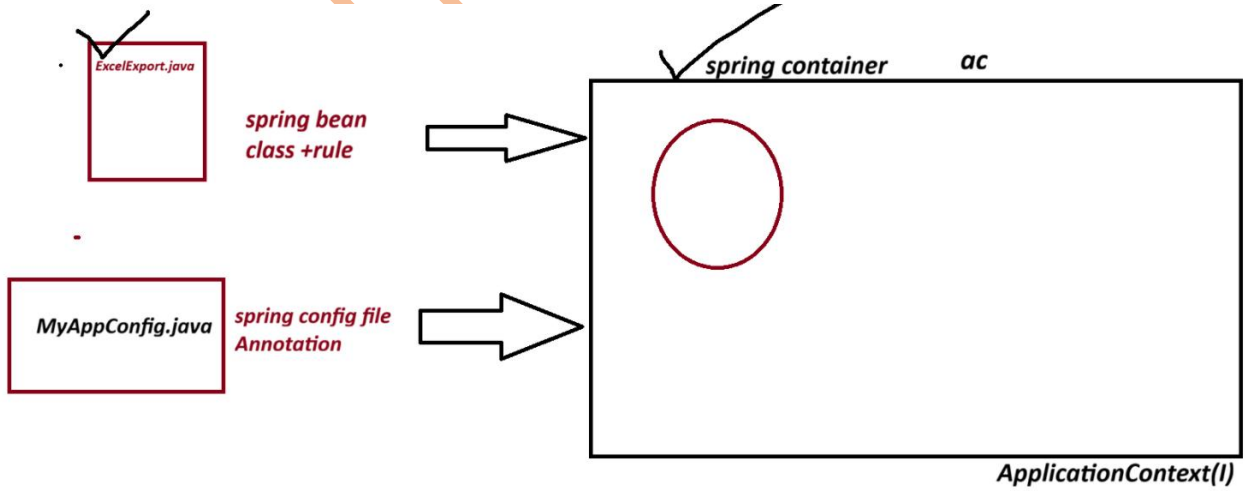
```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
import org.springframework.context.ApplicationContext;


public class Test {
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(MyAppConfig.class);
      // Find classes from a package(basePackage)
//ac.scan("com.app");
      // object creation
     //ac.refresh();



      ExcelExport ee = ac.getBean("excel",ExcelExport.class);
      System.out.println(ee);
}
}
```

output

ExcelExport [exportName=TEST, fileExt=.csv]

Diagram



==========

session 11

========

Spring Annotation Configuration:

1. @Component : create Object to given class

2. @Value : To provide data (Field Injection) to variable

3. @ComponentScan : To find classes using basepakage.

==========Prorperties File==========

_____.properties

=> This is also caleld as key-val pair input as Text format(String)

=> This file is used to provide setup data to application.

=> Here, we can read data from properties file into variable using @Value("${key}")

=>Programmer has to define and load Properties file.

Use @PropertySource annotatio to load file, into spring container.

=> container creates Environment(I) Object (internally used StandardEnvironment(c)).

=> we can read data using @Value("${keyName}")

=> if we define duplicate key with different value (not recomened then last combination is taken into Container)

==========

myapp.properties

#Symbol allowed for comment '#'

#key=value

#Symbol allwed to write keyName (dot(.), dash(-), underscore(_))

my.driver=oracle

my.url=oracle-jdbc

my.username=system

my.password=root

my.port=3306

my.active=fALSE

==========

my.port=4444 will be printed

==========

=> For boolean datatype valid input are : false and true

(case-insensitive, False,FASLE,FalSE, True,TRue etc)

========

Project Name: database-connectionS11

Example

code

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>database-connectionS11</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
public class Test {

public static void main(String[] args) {
ApplicationContext ac=new AnnotationConfigApplicationContext(MyAppConfig.class);
DatabaseCon obj = ac.getBean("databaseCon",DatabaseCon.class);
System.out.println(obj);
}

}
```

DatabaseCon.java

```
package com.app;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

@Component
public class DatabaseCon {

@Value("${my.db}")
private String driver;

@Value("${my.url}")
private String url;

@Value("${my.username}")
private String username;

@Value("${my.password}")
private String password;
```

```
@Value("${my.port}")
private int port;

@Value("${my.enable}")
private boolean active;

@Override
public String toString() {
return "DatabaseCon [driver=" + driver + ", url=" + url + ", username=" + username + ", password=" + password
+ ", port=" + port + ", active=" + active + "]";
}



}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.PropertySource;

@ComponentScan("com.app")
@PropertySource("classpath:myapp.properties")
public class MyAppConfig {


}
```

---

myapp.properties

```
# comment line
# symbol allowed to write keyName (dot . dash - underscore _)
#key=value
my.db=oracle
```

my.url=oracle-jdbc

my.username=indiafreeinternship

my.password=root

my.port=3306

#my.port=3345

#my.port=4444

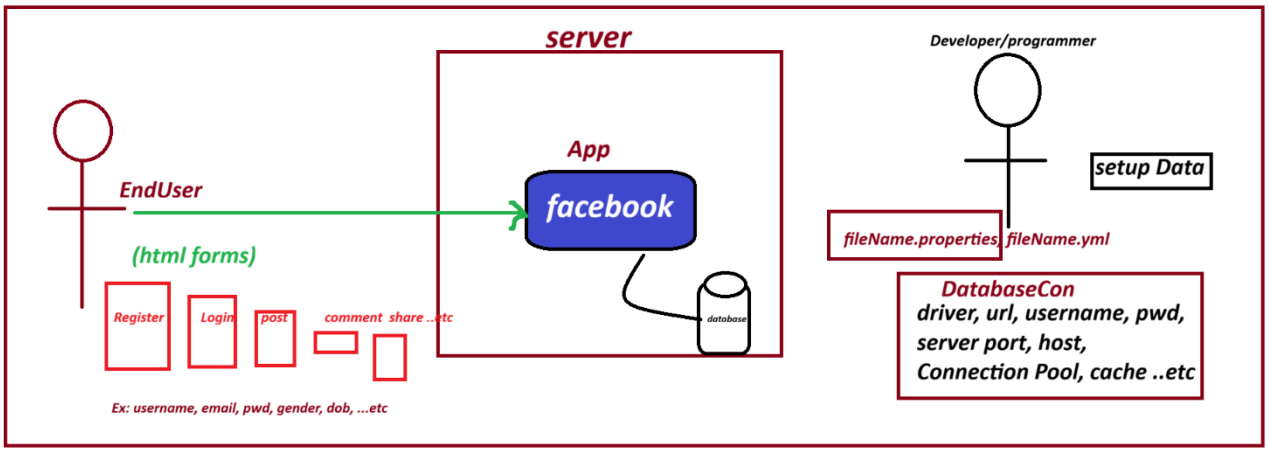
my.enable=tRUe

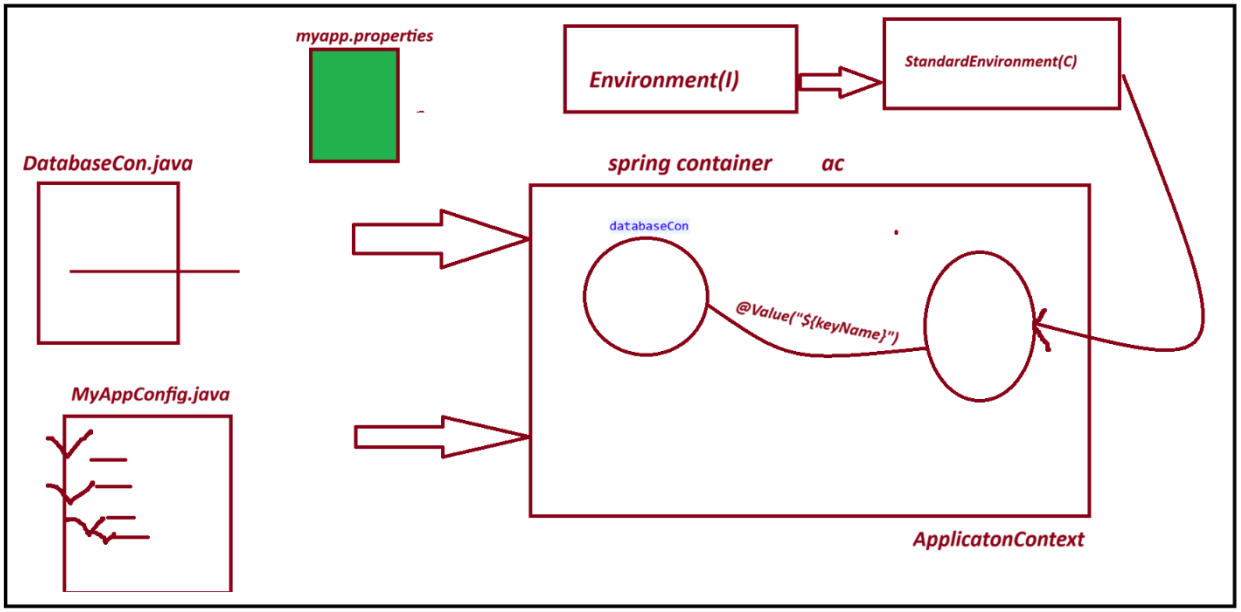
----

output

DatabaseCon [driver=oracle, url=oracle-jdbc, username=indiafreeinternship, password=root, port=4444, active=true]

Diagram





session 12

=======

Spring Annotation configuration

@Component Annotation : To create Object

@Value : Direct value to a variable / Read data from properties

@ComponentScan : back-Package to scan classes

@PropertySource : To Load Properties File

Autowiring

@Autowired Annotation

=> It is a process of Linking objects based on Association Mapping (HAS-A)

=> @Autowired Annotation is used. So, that container can link them.

=>Autowired Annotation Injects Dependecny Beans into Dependent bean.

=> In Case of XML Configuration <ref/> tag is used to link objects.

==========

1. @Autowired will search for child class object.

2. If one child object is found then it is injected.

3. If Zero child object are found then throw exception NoSuchBeanDefinitionException, message like:

At least 1 bean required

==========

---------code----------

project name : AutowiredAnnoS12

==========

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>AutowiredAnnoS12</artifactId>
```

```
<version>1.0</version>
<properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>


</project>
```

---

2. Spring bean

EmployeeDao.java

```
package com.app;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

//@Component("edao")
@Component("edao")
public class EmployeeDao {
```

```
@Value("ORACLE")
private String dbName;

@Override
public String toString() {
return "EmployeeDao [dbName=" + dbName + "]";
}

}
```

---

3. spring bean

EmployeeService.java

```
package com.app;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
//ctrl+shift+O
@Component("esobj")
public class EmployeeService {

//@Autowired(required=true)
@Autowired(required=false)
private EmployeeDao dao; // HAS-A

@Override
public String toString() {
return "EmployeeService [dao=" + dao + "]";
}
```

```
}
```

AppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

Test.java

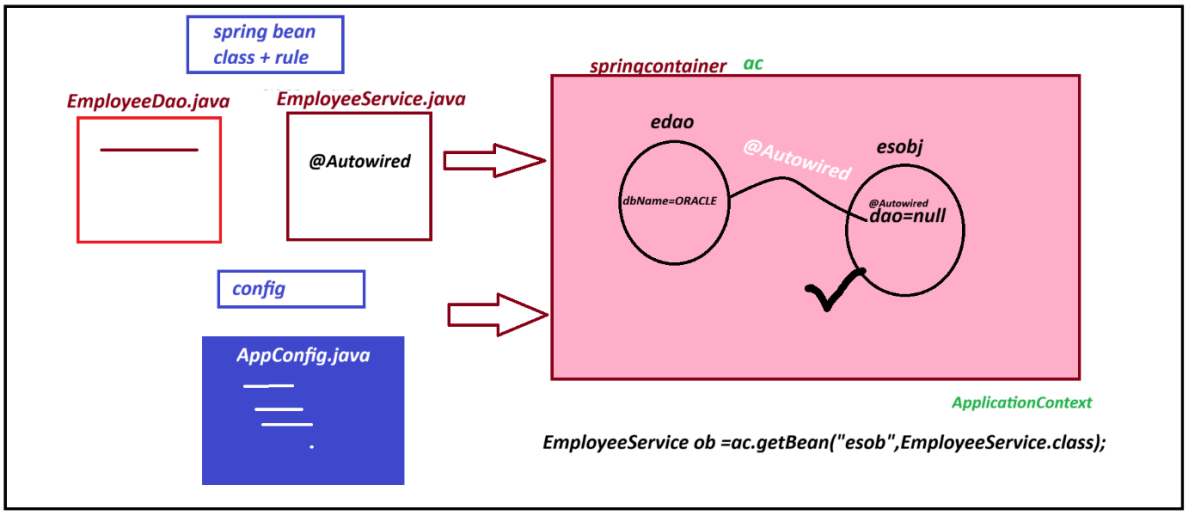
```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

public static void main(String[] args) {
ApplicationContext ac=new AnnotationConfigApplicationContext(AppConfig.class);
EmployeeService ob = ac.getBean("esobj",EmployeeService.class);
System.out.println(ob);

}

}
```

Output

```
EmployeeService [dao=EmployeeDao [dbName=ORACLE]]
```

==========

Diagram



==========

Sessino 13

Spring Bean -LifeCycle Methods

==========

LifeCycle Methods : If a method is called by container/runtime in betwenn Object creation and destroy.

Spring F/W : 2 LifeCycle Methods (they are optional)

1. init-method : called after creating object

2. destroy-method : called before Distorying object

==========

=========init========== ========destroy=====

File(data) -----> open file----------> Read data--------variable-> close File

==========

We can implement LifeCycle Methods in 3 way

1. xml cofiguration

<bean ..... init-method="" destroy-method="">

=>we can define methods with any name, using below syntax:

public void <method_name>(){

//logic....

}

2. spring Lifecycle Interface:

=> Spring f/w has given 2 interfaces to implement lifecycle methods.

They are:

A. InitializingBean(I)

afterPropertiesSet()

B. DisposableBean(I)

Destroy()

=> Your Spring bean can implement above interface and override methods.

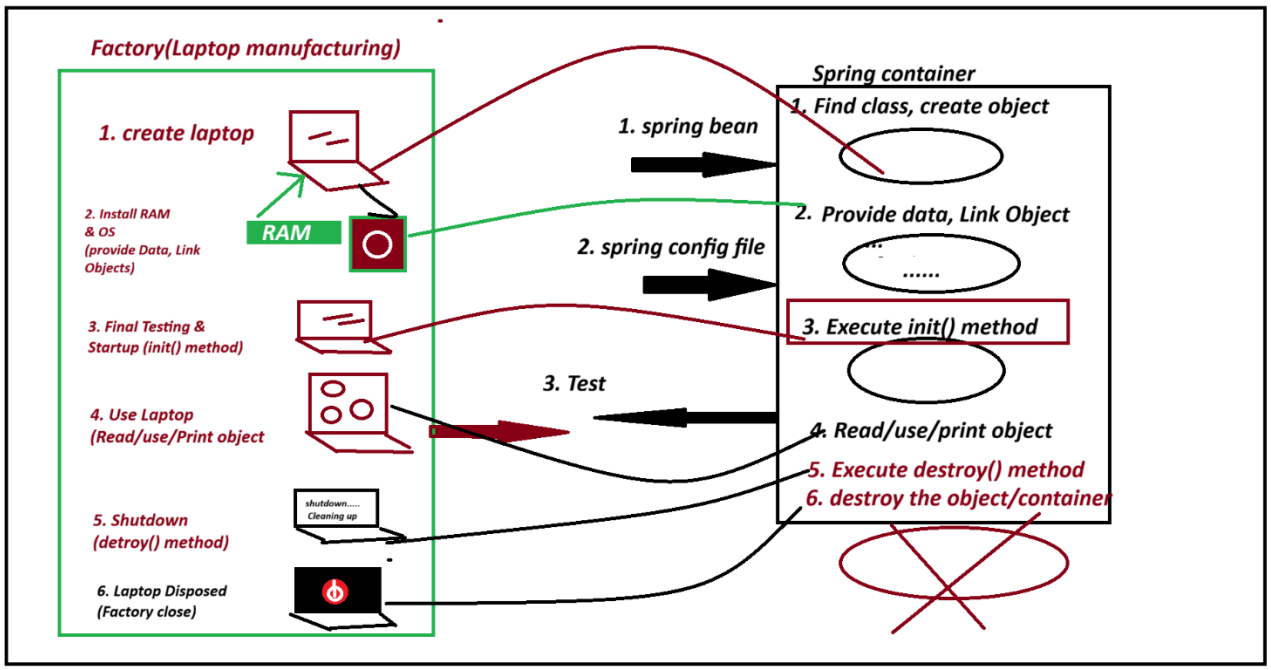
They are called by spring container.

3. JSR-Annotation : These are generic annotation Given by Sun/oracle .

a. **@PostConstruct**

b. **@PreDestroy**

diagram



==========code==========

1. xml cofiguration

<bean ..... init-method="" destroy-method="">

=>we can define methods with any name, using below syntax:

public void <method_name>(){

//logic....

}

==========code==========

project :- SpringBeanLifeCycleXMLS13

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>SpringBeanLifeCycleXMLS13</artifactId>
 <version>0.0.1-SNAPSHOT</version>
 <properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
```

```
</dependencies>
</project>
```

---

spring bean

ExcelExport.java

```
package com.app;

public class ExcelExport {

private String fileName;
private String fileExt;

public ExcelExport() {
super();
System.out.println("from constructor");
}

public String getFileName() {
return fileName;
}

public void setFileName(String fileName) {
this.fileName = fileName;
System.out.println("from setter method");
}

public String getFileExt() {
return fileExt;
}

public void setFileExt(String fileExt) {
this.fileExt = fileExt;
```

```
}

public void setUp(){
System.out.println("3.FROM INIT METHOD");
}

public void clear() {
System.out.println("5.FROM DESTORY METHOD");
}

@Override
public String toString() {
return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}

}
```

2. config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

<bean class="com.app.ExcelExport" id="exObj" init-method="setUp" destroy-method="clear">
<property name="fileName" value="ABC"/>
<property name="fileExt" value="CSV"/>
</bean>

</beans>
```

---

3. Test.java

```
package com.app;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;


public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
ClassPathXmlApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
ac.registerShutdownHook();
//ac.close();
}


}
```

---

output

```
from constructor
from setter method
3.FROM INIT METHOD
FROM MAIN
ExcelExport [fileName=ABC, fileExt=CSV]
5.FROM DESTORY METHOD
```

==========

2. spring Lifecycle Interface:

=> Spring f/w has given 2 interfaces to implement lifecycle methods.

They are:

A. InitializingBean(I)

   afterPropertiesSet()

B. DisposableBean(I)

Destroy()

=> Your Spring bean can implement above interface and override methods.

They are called by spring container.

==========code==========

project Name :SpringBeanLifeCycleInterfaceS13

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>SpringBeanLifeCycleXMLS13</artifactId>
 <version>0.0.1-SNAPSHOT</version>
 <properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

ExcelExport.java

```
package com.app;
```

```
import org.springframework.beans.factory.DisposableBean;
import org.springframework.beans.factory.InitializingBean;

public class ExcelExport implements InitializingBean, DisposableBean{

private String fileName;
private String fileExt;

public ExcelExport() {
super();
System.out.println("from constructor");
}

public String getFileName() {
return fileName;
}

public void setFileName(String fileName) {
this.fileName = fileName;
System.out.println("from setter method");
}

public String getFileExt() {
return fileExt;
}

public void setFileExt(String fileExt) {
this.fileExt = fileExt;
}
```

```
@Override
public String toString() {
return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}
@Override
public void afterPropertiesSet() throws Exception {
System.out.println("FROM INIT METHOD AFTER PROPERTIESSET()");

}

@Override
public void destroy() throws Exception {
System.out.println("from destroy method destroy()");

}



}
===============
config.xml
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

<bean class="com.app.ExcelExport" id="exObj">
<property name="fileName" value="ABC"/>
<property name="fileExt" value="CSV"/>
</bean>

</beans>
```

Test.java

```
package com.app;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
ClassPathXmlApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
ac.registerShutdownHook();
//ac.close();
}

}
```

---

output

```
from constructor
from setter method
FROM INIT METHOD AFTER PROPERTIESSET()
FROM MAIN
ExcelExport [fileName=ABC, fileExt=CSV]
from destroy method destroy()
```

==========

3. JSR-Annotation : These are generic annotation Given by Sun/oracle .

a. **@PostConstruct**
b. **@PreDestroy**

--------code----------

```
 <dependency>
<groupId>jakarta.annotation</groupId>
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>
```

---

project name : SpringBeanLifeCycleAnnotationS13

code

ExcelExport.java

```
package com.app;



import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

import jakarta.annotation.PostConstruct;
import jakarta.annotation.PreDestroy;

@Component("exObj")
public class ExcelExport {

@Value("SAMPLE")
private String fileName;
@Value(".csv")
private String fileExt;

public ExcelExport() {
super();
```

```
System.out.println("from constructor");
}

@Override
public String toString() {return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}



@PostConstruct
public void setUpA() {
System.out.println("FROM INIT METHOD");
}

@PreDestroy
public void clearB() {
System.out.println("FROM DESTROY METHOD");
}

}
```

Test.java

```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
```

```
ac.registerShutdownHook();
//ac.close();
}

}
```

AppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SpringBeanLifeCycleXMLS13</artifactId>
<version>0.0.1-SNAPSHOT</version>
<properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
```

https://mvnrepository.com/artifact/org.springframework/spring-context -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

<dependency>

<groupId>jakarta.annotation</groupId>

<artifactId>jakarta.annotation-api</artifactId>

<version>2.0.0</version>

</dependency>

</dependencies>

</project>

---

output

---

from constructor

FROM INIT METHOD

FROM MAIN

ExcelExport [fileName=SAMPLE, fileExt=.csv]

FROM DESTROY METHOD

==========

session 14

==========

Spring - Java Configuration

==========

XML Configuration -- <bean> <property> <value> .. etc

Annotation configuration -- @Component, @Value, @Autowired .etc

java configuration -@Configuration @Bean

==========

java configuration -@Configuration @Bean

1. Define one public class with any name (ex MyApp.java)

2. Apply @Configuration annother over class

3. For one object = define one method

```
@Bean
public PdfExport pObj(){
PdfExport p = new PdfExport();
p.setFileName("sbms");
p.setFileAuth("IndiaFreeInternship");
return p;

}
```

4. Add @Bean over method

---

1. Consider given class (spring bean) is a pre-defined class.

  can we use Annotation config?

  => No. We can't Annotation configuration.

      we can use annotation configuration we have source code only.

(_____.java) file

2. Can we xml/java configuration to define one bean for pre-defined class?

   yes Both support any type of class of configuration (pre-defined and user-defined)

3. what is the mean of @Configuration and @Bean?

    @Configuration says "i am java config file"

 @Bean - please create one object inside spring container

4. what is the diff b/w @Component and @Bean?

 @Component - used for programmer defined

 @Bean - Mainly for pre-defined class (support both pre-defined and user defined)

5. why we use Annotation configuration mostly

execution fast (short code)

in java configuration (lengthy code and slow execution)

==========

example code

PdfExportJavaConfigS14

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>PdfExportJavaConfigS14</artifactId>
 <version>1.0</version>
 <properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
<dependency>
<groupId>jakarta.annotation</groupId>
```

```
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>

<!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-jdbc -->
<dependency>
   <groupId>org.springframework</groupId>
   <artifactId>spring-jdbc</artifactId>
   <version>7.0.7</version>
   <scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Test.java

```
package com.app.test;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.app.bean.PdfExport;
import com.app.config.AppConfig;

public class Test {
@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
PdfExport ob = ac.getBean("pObj",PdfExport.class);

System.out.println(ob);

}
```

```
}
```

PdfExport.java

```
package com.app.bean;

public class PdfExport {

private String fileName;
private String fileAuth;
public String getFileName() {
return fileName;
}
public void setFileName(String fileName) {
this.fileName = fileName;
}
public String getFileAuth() {
return fileAuth;
}
public void setFileAuth(String fileAuth) {
this.fileAuth = fileAuth;
}
@Override
public String toString() {
return "PdfExport [fileName=" + fileName + ", fileAuth=" + fileAuth + "]";
}



}
```

AppConfig.java

```
package com.app.config;
```

```
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

import com.app.bean.PdfExport;

@Configuration
public class AppConfig {

// user defined
@Bean
public PdfExport pObj(){
PdfExport p = new PdfExport();
p.setFileName("sbms");
p.setFileAuth("IndiaFreeInternship");
return p;

}

}
```

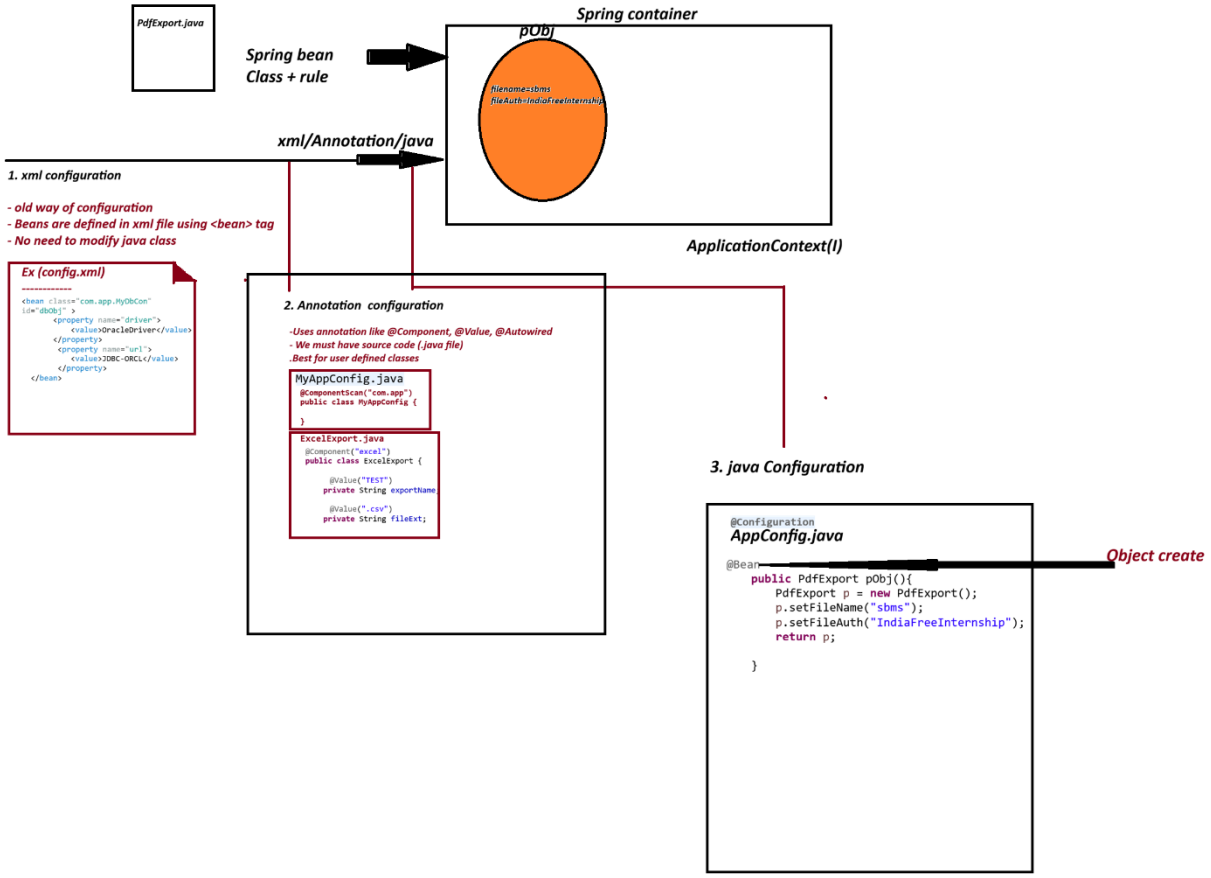
---

output

PdfExport [fileName=sbms, fileAuth=IndiaFreeInternship]

Diagram

==========



Point	xml configuration	Annotation Configuration	Java Configuration
How to define Bean	<bean> tag in XML file	@Component, @Value, @Autowired	@Configuration class with @Bean tag
Need of Source Code	No	Yes(must have .java file)	No (Work for both)
Supports	Both pre-defined and user-defined classes	Only User-defined classes	Both pre-defined and user-defined classes
Multiple beans	Yes	Yes	Yes(easier)
Readability	Medium	High	High
Moder Approad	No	Yes(most used)	Yes

PRE-DEFINED CLASS

classes which are already available in libraries or framework.

we don't have source code. (.java)

ex:
JdbcTemplate
DriverManagerDataSource
RestTemplate
ArrayList(java API)

We Cannot apply @Component
(Cannot modify source code)

USER-DEFINED CLASS

Classes which are creatd by Programmer
We have source code (.java)

Ex
Student.java
Employee.java
Product.java
PdfExport.java

We can apply @Component
(We own the source code)

session 15

=========

Java Configuration : Working with properties file

Lombok API

==========

Java Configuration : Working with properties file

=> properties file holds data in key=val

=> This is used to provide application setup data at runtime.

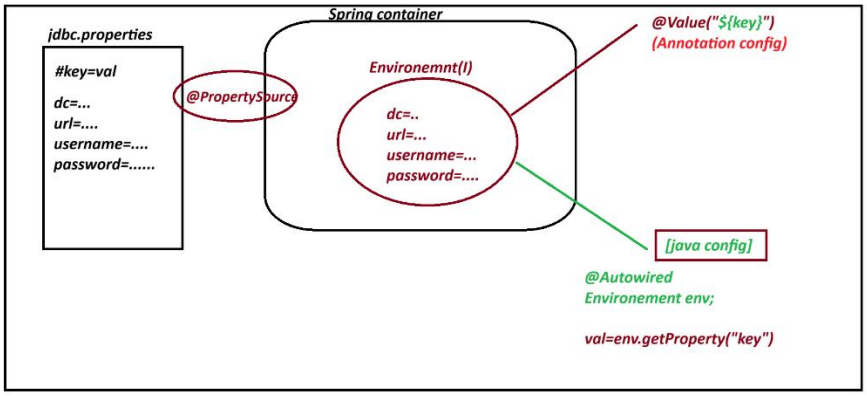
ex : Jdbc Connection details, ORM details, Email Config, Security View Resolvers ..etc

=> .properties file can be loaded using annotation @PropertySource

ex : @PropertySource("classpath:jdbc.properties")

=> Spring Container all these details into Environment(I) memory spring f/w has given impl class : StandardEnvironment(c)

=> This Environment(I) can be read from container to app using @Autowired.



Exmaple code

Dbcon-java-config-propertiesS15

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>Dbcon-java-config-propertiesS15</artifactId>
 <version>1.0</version>
 <properties>
```

```
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>
 <dependencies>
 <dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
<dependency>
<groupId>jakarta.annotation</groupId>
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>

 </dependencies>
</project>
```

MyDbConnection.java

```
package com.app.bean;


public class MyDbConnection {


private String driver;
private String url;
private String username;
private String password;



public MyDbConnection() {
```

```
super();
// TODO Auto-generated constructor stub
}
public String getDriver() {
return driver;
}
public void setDriver(String driver) {
this.driver = driver;
}
public String getUrl() {
return url;
}
public void setUrl(String url) {
this.url = url;
}
public String getUsername() {
return username;
}
public void setUsername(String username) {
this.username = username;
}
public String getPassword() {
return password;
}
public void setPassword(String password) {
this.password = password;
}
@Override
public String toString() {
return "MyDbConnection [driver=" + driver + ", url=" + url + ", username=" + username + ", password=" +
password+ "]";
}
```

```
}
```

AppConfig.java

```
package com.app.config;

import org.jspecify.annotations.Nullable;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.context.annotation.PropertySource;
import org.springframework.core.env.Environment;

import com.app.bean.MyDbConnection;

@Configuration
@PropertySource("classpath:jdbc.properties")
public class AppConfig {

@Autowired
Environment env;

@Bean
public MyDbConnection dbObj() {
MyDbConnection d = new MyDbConnection();
d.setDriver(env.getProperty("my.app.driver-class"));
d.setUrl(env.getProperty("my.app.url"));
d.setUsername(env.getProperty("my.app.username"));
d.setPassword(env.getProperty("my.app.password"));
return d;
}
```

Test.java

```
package com.app.test;


import org.springframework.context.annotation.AnnotationConfigApplicationContext;


import com.app.bean.MyDbConnection;
import com.app.config.AppConfig;


public class Test {


@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
MyDbConnection ob = ac.getBean("dbObj",MyDbConnection.class);
System.out.println(ob);
}


}
```

---

jdbc.properties

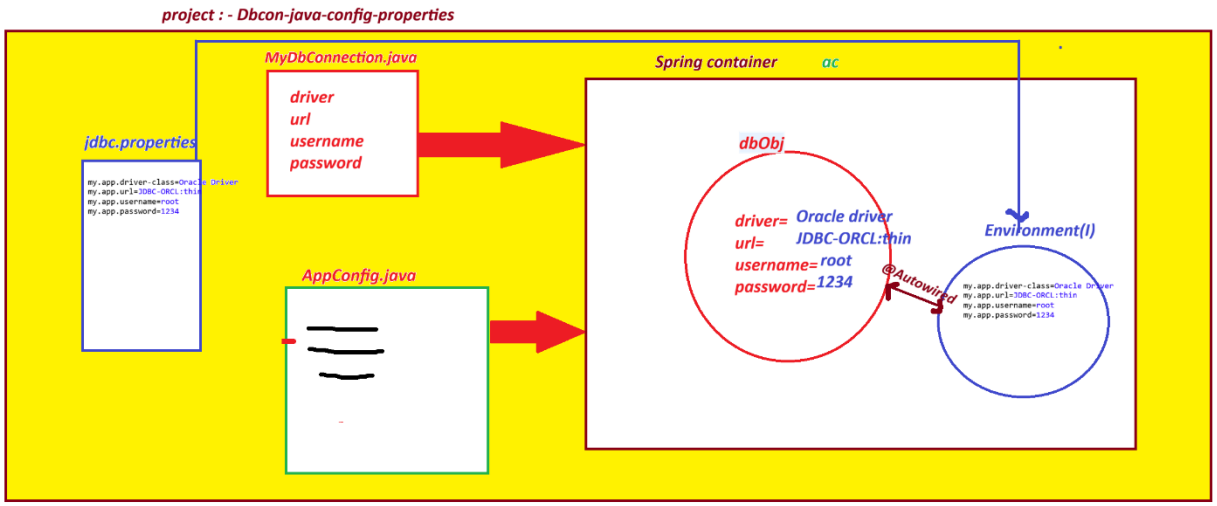
```
# key name can have A-Z a-z dash(-) dot(.) underscore(_)
my.app.driver-class=Oracle Driver
my.app.url=JDBC-ORCL:thin
my.app.username=root
my.app.password=1234


}
```

---

output

MyDbConnection [driver=Oracle Driver, url=JDBC-ORCL:thin, username=root, password=1234]

Diagram



==========

PROJECT Lombok API

==========

=> It is open source Java API, used to generate code if we apply annotations

=> Lombok can generate code like

- Constructor (default, Parameterized)
- setter/getter methods
- toString()
- hashCode()
- equals()

==========

Annotation	Work
@Getter	Generate getter()
@Setter	Generate setter()
@ToString	toString()
@NoArgsConstructor	Default constructor
@AllArgsConstructor	Parameterzed constructor
@Data	Getter+Setter+toString+equals+hashCode

@Builder **Builder Patter implement**

@RequiredArgsConstructor **Required fields constructor**

----

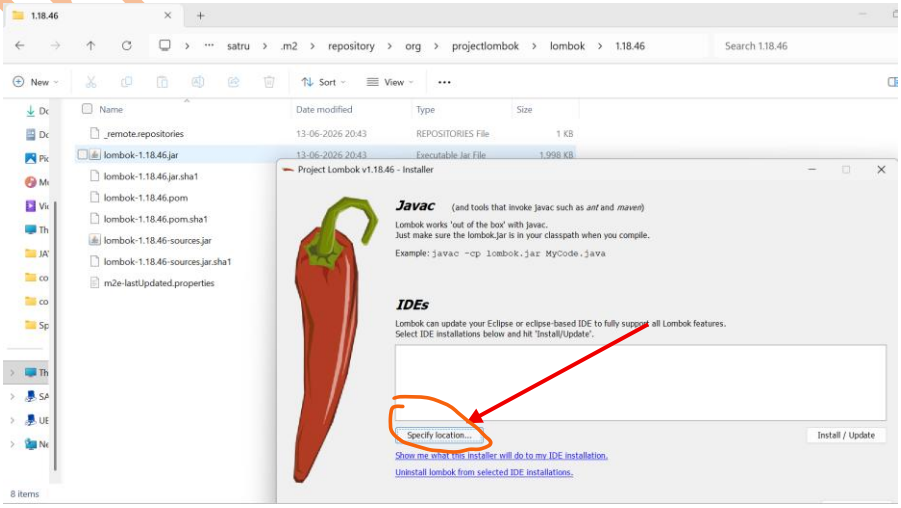
example:

```
import lombok.Getter;
import lombok.NoArgsConstructor;
import lombok.Setter;
import lombok.ToString;
@NoArgsConstructor
@Setter
@Getter
@ToString
public class MyDbConnection {

private String driver;
private String url;
private String username;
private String password;

}
```

==========

Go for Lombok Installation

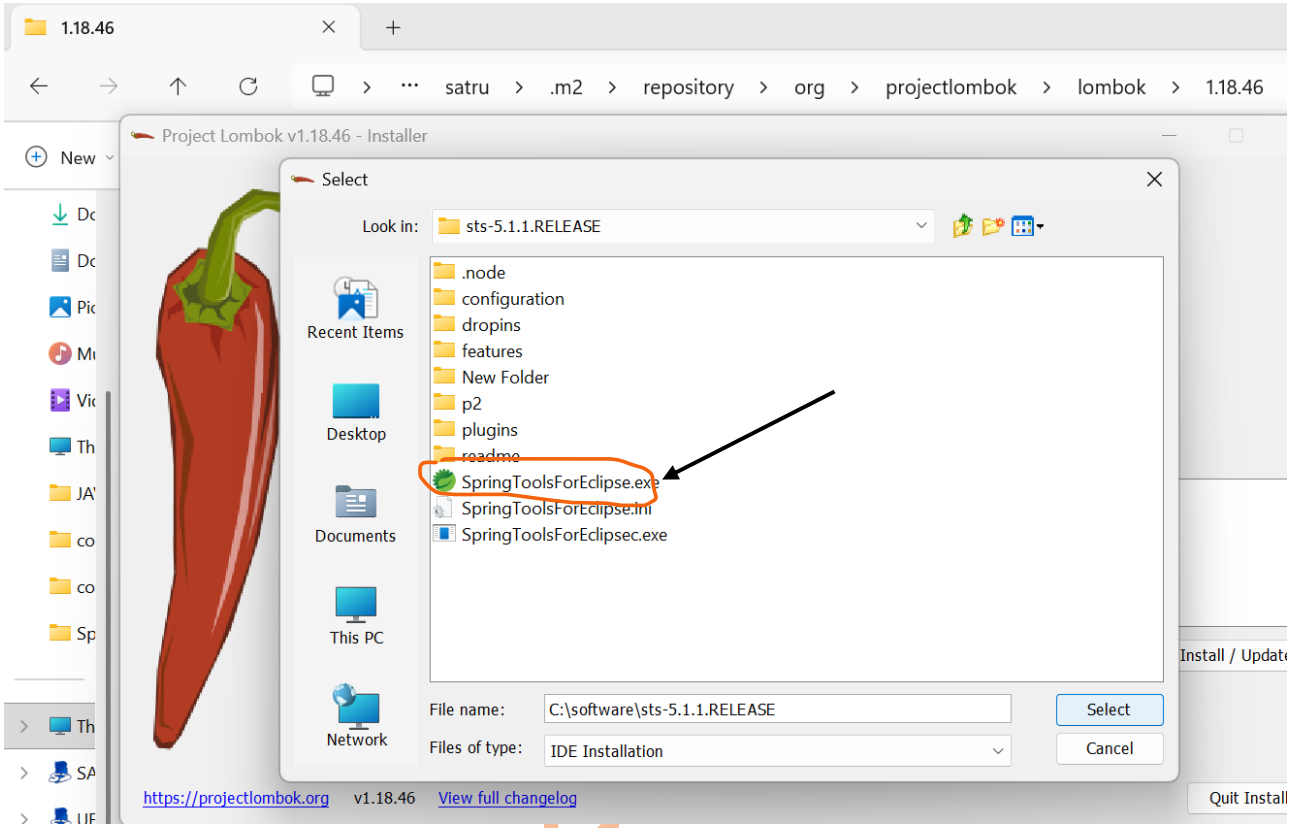
1. **C:\Users\satru\.m2\repository\org\projectlombok\lombok\1.18.46**



2. double click : lombok-1.18.46

3. wait sometime to detect STS IDE location

If not click on specifiy location and select sts IDE location



4. click on Install/Update > quite.

5. Open Again

==========

session 16

==========

Spring bean scope:

Scope: - Life time of object present in memory/ How Long object exist in memory.

Spring bean Scope: When bean(object) is created, how long it is present in container.

1. singleton

2. prototype

3. request

4. session

1. signleton(default) : for every bean this is default scope. One object is created per configuration.

It returns single object even if we acess it multiple times.

2. prototype : In prototype scope, a new bean instance is creatd every time the bean is requested from the container.

3. request : web app container new object when a new request;

4. session : web app container creates object when new session is created (Login-Logout) and object is removed when session is destoryed.

---

1. spring xml Configuration:

Ex program

project name: bean-scope-s16

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>bean-scope-s16</artifactId>
 <version>1.0</version>
 <properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>

<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
```

```
<scope>compile</scope>
</dependency>

<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<version>1.18.46</version>
<scope>provided</scope>
</dependency>
</dependencies>

</project>
```

Product.java

```
package com.app;

import lombok.Getter;
import lombok.Setter;
import lombok.ToString;

@Setter
@Getter
@ToString
public class Product {

private int pid;
private String pcode;




}
```

Test.java

```
package com.app;


import org.springframework.context.support.ClassPathXmlApplicationContext;


public class Test {


@SuppressWarnings("resource")
public static void main(String[] args) {
ClassPathXmlApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");
/*singleton
Product ob1 = ac.getBean("pobj",Product.class);
Product ob2 = ac.getBean("pobj",Product.class);
Product ob3 = ac.getBean("pobj",Product.class);



System.out.println(ob2==ob3);
 */
Product p1 = ac.getBean("pobj",Product.class);
Product p2 = ac.getBean("pobj",Product.class);
Product p3 = ac.getBean("pobj",Product.class);
System.out.println(p1==p2);
}
}
```

---

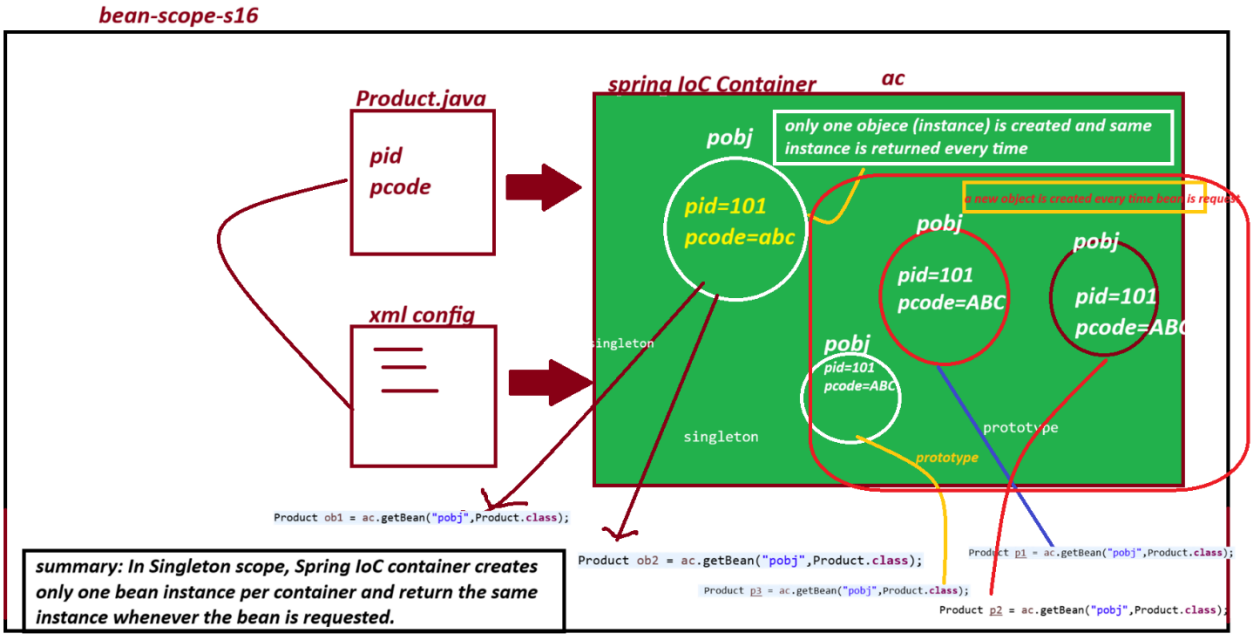
config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
 <bean class="com.app.Product" id="pobj" scope="prototype">
 <property name="pid" value
="101"/>
 <property name="pcode" value="ABC"/>

 </bean>
</beans>
```

==========

xml diagram



---

2. Spring Annotation Configuration

example

project name : bean-scope-annotation-s16

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
```

```
 <groupId>com.app</groupId>
 <artifactId>bean-scope-annotation-s16</artifactId>
 <version>1.0</version>
 <properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>

<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>

<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<version>1.18.46</version>
<scope>provided</scope>
</dependency>
</dependencies>
</project>
```

Product.java

```
package com.app;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.context.annotation.Scope;
```

```
import org.springframework.stereotype.Component;

import lombok.Getter;
import lombok.Setter;
import lombok.ToString;

@Getter
@Setter
@ToString
@Component("pobj")
//@Scope("singleton")
@Scope("prototype")
public class Product {

@Value("101")
private int pid;
@Value("ABC")
private String pcode;
}
```

Test.java

```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
Product p1 = ac.getBean("pobj",Product.class);
Product p2 = ac.getBean("pobj",Product.class);
```

System.out.println(p1==p2);

}

}

---

config.java

package com.app;

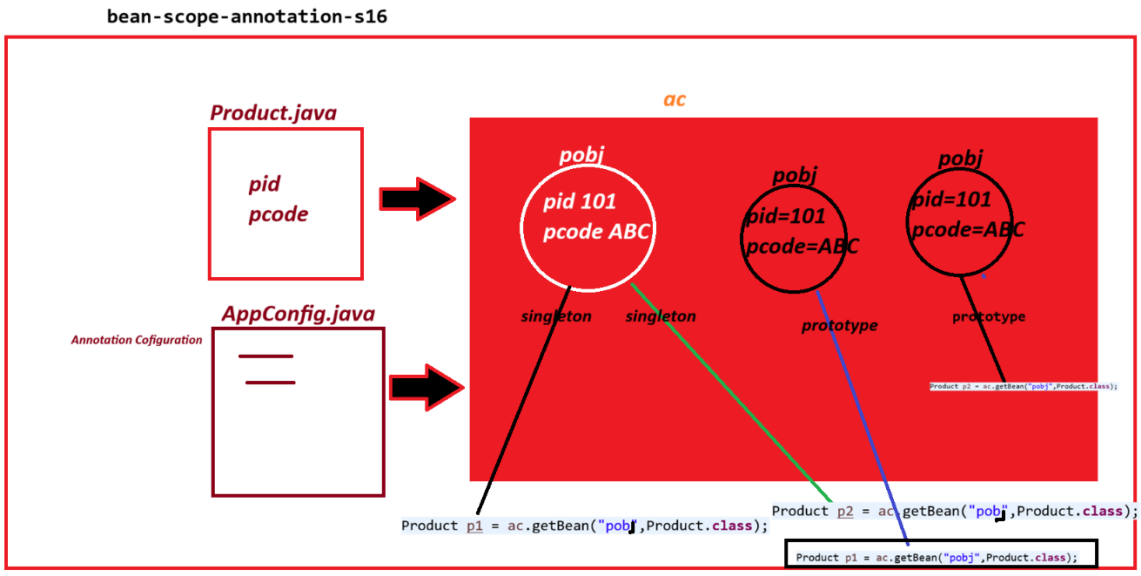
import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")

public class AppConfig {

}

digram



==========

3. Spring java configuration

example bean-scope-java-config-s16

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>bean-scope-java-config-s16</artifactId>
 <version>1.0</version>
 <properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>

<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>

<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<version>1.18.46</version>
<scope>provided</scope>
</dependency>
</dependencies>
</project>
```

Product.java

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>bean-scope-java-config-s16</artifactId>
 <version>1.0</version>
 <properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>

<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>

<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<version>1.18.46</version>
<scope>provided</scope>
</dependency>
</dependencies>
</project>
```

AppConfig.java

```
package com.app;


import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.context.annotation.Scope;


@Configuration
public class AppConfig {


@Bean
//@Scope("singleton")
@Scope("prototype")
public Product pobj(){
Product p = new Product();
p.setPid(101);
p.setPcode("ABC");
return p;
}
}
```

Test.java

```
package com.app;


import org.springframework.context.annotation.AnnotationConfigApplicationContext;


public class Test {
@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
Product p1 = ac.getBean("pobj",Product.class);
Product p2 = ac.getBean("pobj",Product.class);
                                                                    System.out.prin
tln(p1==p2);
```

}

}

ouput

for prototype

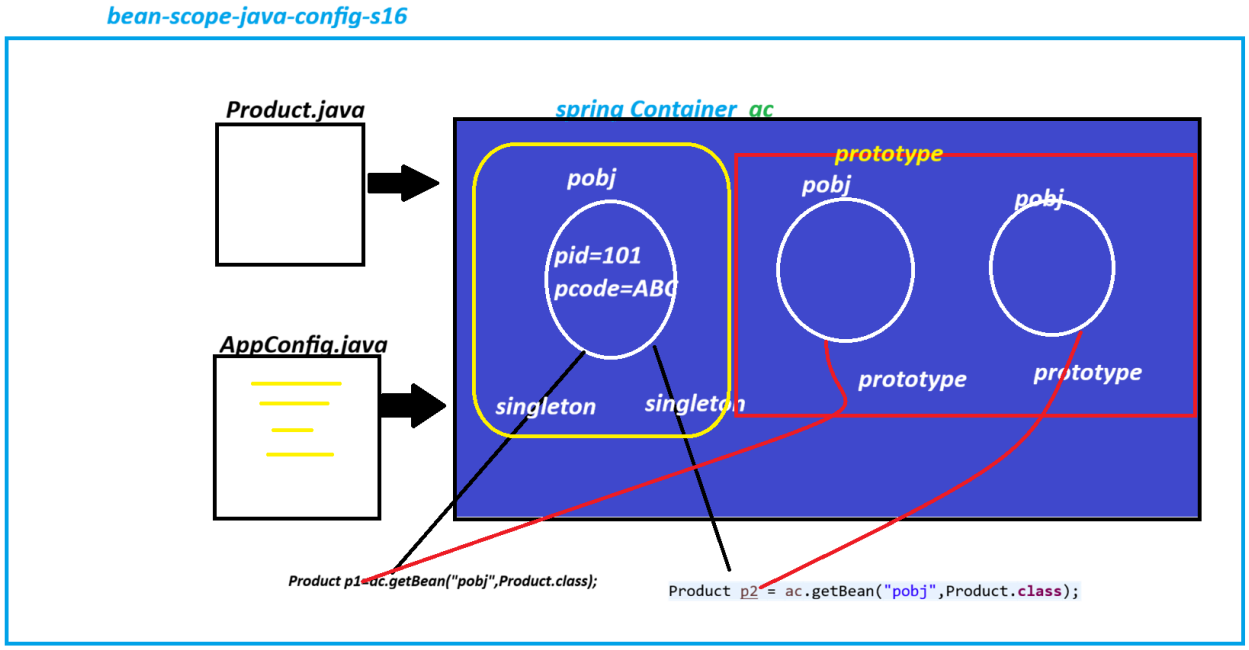
FROM CONSTRUCTOR

FROM CONSTRUCTOR

false

==========

diagram



========

Session 17

==========

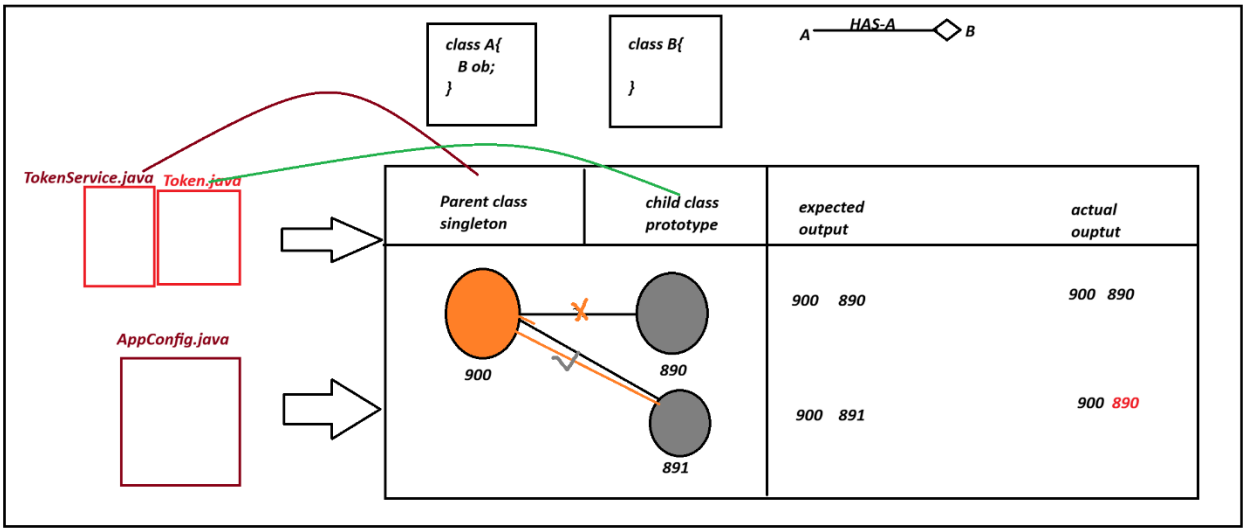
Spring Bean Scope - Lookup Method Injection (LMI)

problem statement:

If two Spring beans(classes) are connected using Rerence Type(HAS-A) and child class(independent) scope is prototype and parent class(Dependent) scope is singleton. Then they may not get linked as expected .

or Always parent links to first child object.

diagram



---

code

---

project name bean-scope-LMI-s17

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.app</groupId>
 <artifactId>bean-scope-LMI-s17</artifactId>
 <version>1.0</version>
  <properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
```

```
<dependencies>

<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>

<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<version>1.18.46</version>
<scope>provided</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

Token.java

```
package com.app.bean;

import java.util.Random;

import org.springframework.context.annotation.Scope;
import org.springframework.stereotype.Component;

@Component
@Scope("prototype")
public class Token {

private int token;
```

```
public Token() {token = new Random().nextInt(999);
}

@Override
public String toString() {
return "Token [token=" + token + "]";
}



}
```

Spring bean

TokenService.java

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.annotation.Scope;
import org.springframework.stereotype.Component;

@Component
@Scope("singleton")
public class TokenService {

@Autowired
private Token tokenOb;

public Token getTekenOb() {
return tokenOb;
}

@Override
```

```
public String toString() {
return "TokenService [tokenOb=" + tokenOb + "]";
}

}
```

==========

AppConfig.java

```
package com.app.config;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

Test.java

```
package com.app.test;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.app.bean.TokenService;
import com.app.config.AppConfig;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
TokenService ts1 = ac.getBean("tokenService",TokenService.class);
System.out.println(ts1);
System.out.println("Token Serive HS : "+ ts1.hashCode());
```

```
System.out.println("Token HS :"+ ts1.getTekenOb().hashCode());


TokenService ts2 = ac.getBean("tokenService",TokenService.class);
System.out.println(ts2);
System.out.println("Token Serive HS : "+ ts2.hashCode());
System.out.println("Token HS :"+ ts2.getTekenOb().hashCode());
}


}
```

==========

ouput

TokenService [tokenOb=Token [token=507]]

Token Serive HS : 178049969

Token HS :333683827

TokenService [tokenOb=Token [token=507]]

Token Serive HS : 178049969

Token HS :333683827

==========

Solution:

This solution is given by spring f/w. porgrammer need not to implement any logic. But need to provide some details to container.

-> code Modifications come at parent class only.

1. define one public method with empty body.
2. Return type of such method should be child type
3. and return null statement
4. Add one annotation over method @Lookup
5. class this method some where is code (ex: getter method)

--- solution code--------

TokenService.java

```
package com.app.bean;
```

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Lookup;
import org.springframework.context.annotation.Scope;
import org.springframework.stereotype.Component;

@Component
@Scope("singleton")
public class TokenService {

@Autowired
private Token tokenOb;

// problem solution
@Lookup
public Token linkNewChildObj() {
return null;
}

public Token getTekenOb() {

this.tokenOb = linkNewChildObj();
return tokenOb;
}

@Override
public String toString() {
return "TokenService [tokenOb=" + tokenOb + "]";
}

}
```

==========

solution output

TokenService [tokenOb=Token [token=997]]

Token Serive HS : 130668770

Token HS :330739404

TokenService [tokenOb=Token [token=349]]

Token Serive HS : 130668770

Token HS :361398902

==========

Circular Dependency

==========

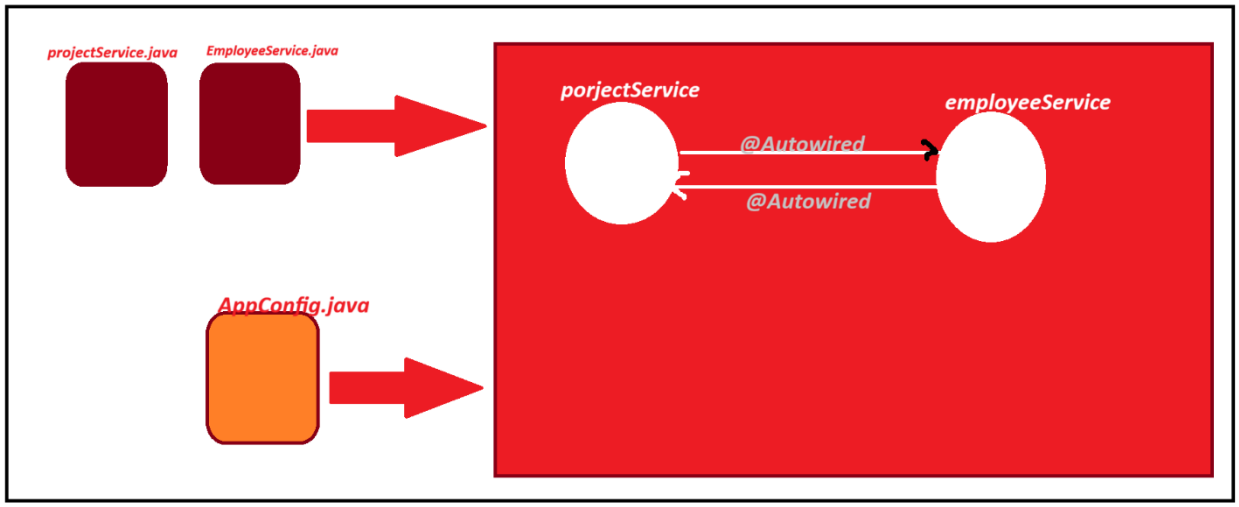
=> If two classess are connected using Reference (HAS-A) Relation then objects are created using default constructors first and linked using set methods one after another.

such process is called as circular Dependency and handled by Spring container.

:- do not override toString() method in both classes.

if we do, then toString() method are called in loop from both classses then results :

java.lang.StackOverFlowError




```
@Component
public class A{
  @Autowired
```

```
  private B b;
}
@Component
public class B{
@Autowired
private A a;
}
```

---

project name :

circular-dependency-s17

pom.xml

some of above

EmployeeService.java

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component
public class EmployeeService {

@Autowired
private ProjectService pservice;

}
```

ProjectService.java

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
```

```
@Component
public class ProjectService {

@Autowired
private EmployeeService eService;

@Override
public String toString() {
return "ProjectService [eService=" + eService + "]";
}



}
```

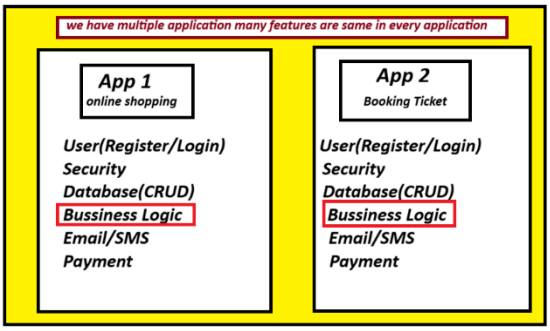
AppConfig.java

```
package com.app.config;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

Test.java

```
package com.app.test;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.app.bean.ProjectService;
import com.app.config.AppConfig;
```

```
public class Test {


@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
ProjectService ps = ac.getBean("projectService",ProjectService.class);
System.out.println(ps);
}


}
==============
output
------------
ProjectService [eService=com.app.bean.EmployeeService@2e4b8173]
====================
session 18 (2. SPRING BOOT)
====================
Spring Framework
Spring Boot
---------------
Spring Framework:
=> It is mainly used to devlop web/webservices applcation.
=> Here, programmer has to define configuration [input to spring container.] which is very lengthy and started from line 1.


diagram
----------
```



Disadvantage

==========

-> many application have some features.

-> we have to write same code again and again.

-> It increase development time.

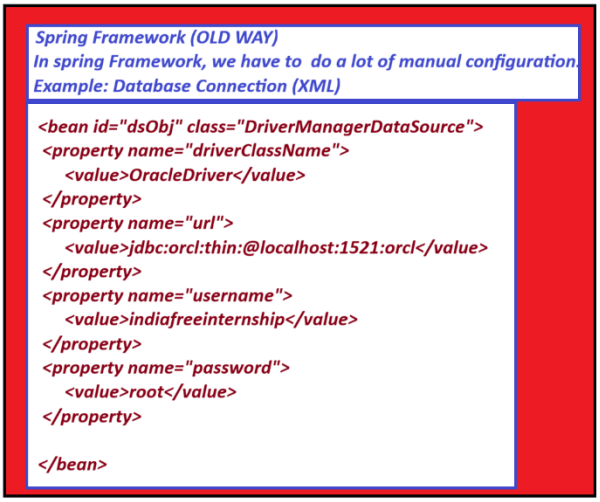
-> It is boring and error-prone.

-> we need an easier and faster solution.

Spring Framework (OLD WAY)

```
<bean id="dsObj" class="DriverManagerDataSource">
 <property name="driverClassName">
    <value>OracleDriver</value>
 </property>
 <property name="url">
    <value>jdbc:orcl:thin:@localhost:1521:orcl</value>
 </property>
 <property name="username">
    <value>indiafreeinternship</value>
 </property>
 <property name="password">
    <value>root</value>
 </property>

</bean>
```

diagram



->In Spring framework, we need to write a lot of XML configuration.

-> we have to configure every component manually.

-> for example : database , Security, Transaction, etc..

-> So much configuration just for one connection!

-> Development becomes slow and comples.

==========

Spring Boot :-

-Spring boot is a spring based framework that provides Auto Configuration, Starter Dependencies and Embedded server suport to

reduce developement time and boilerplate code.

-Spring boot provides all common things in a parent project.

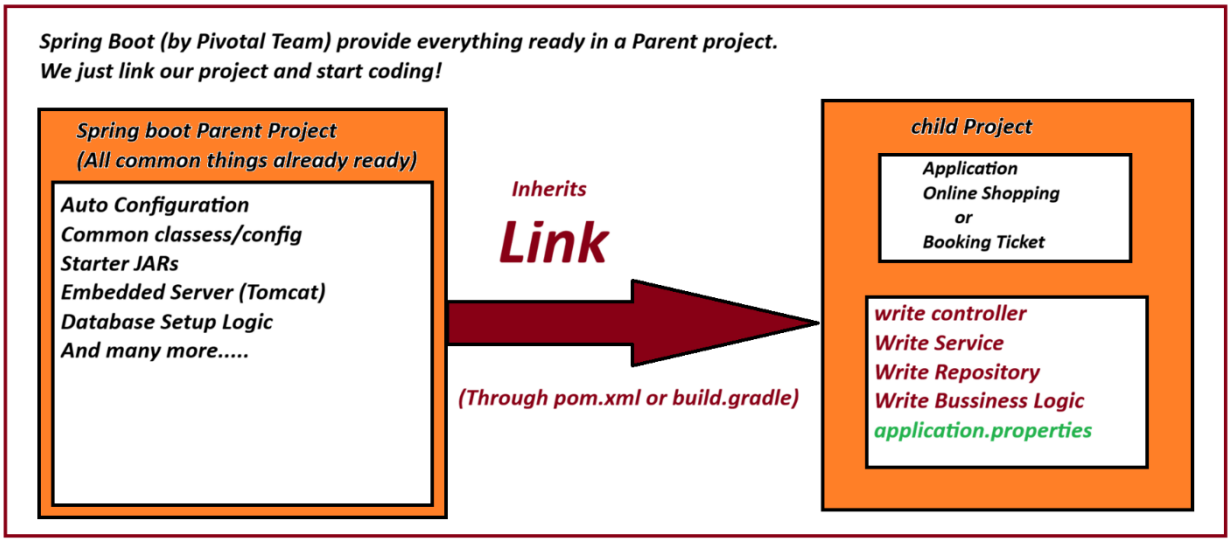
-It includes Auto configuration, Common Classes, Starter JARs, Embedded Server, Database Setup, etc...

-We create our own project (Child project) and link it with parent.

-parent gives all these facilites automatically.

-We just write our bussniess logic and application.properties.

-This saves a lot of time and efforts.



==========

Starter Dependencies (STARTERS)

==========

Starter are ready-made JARs.

They include common code, libraries and configuration.

we just add the required started in pom.xml or build.gradle.

It includes everything we need for that features.

Diagram

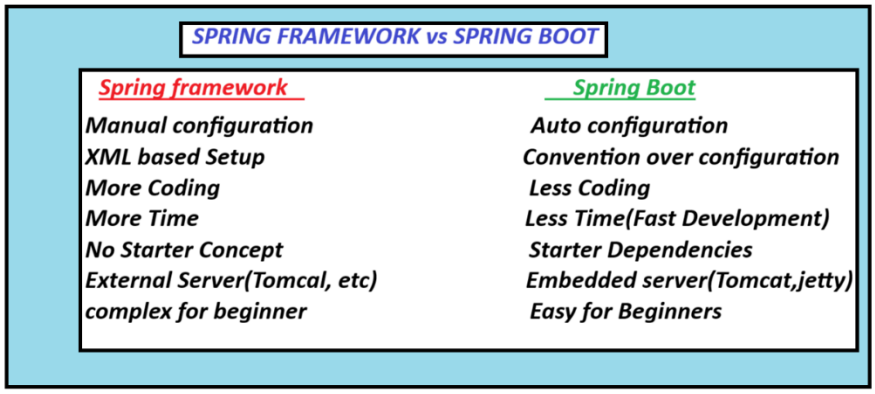
starter Dependencies (Starters)

Starters are pre-build JARs They contain common code and configuration

Starter Name	Use
spring-boot-starter-web	Web(REST APIs, MVC)
spring-boot-starter-jdbc	Database Connectivity (JDBC)
spring-boot-starter-data-jpa	JPA+Hibernate
spring-boot-starter-security	Security(Login, Auth)
spring-boot-starter-mail	Email Sending
spring-boot-starter-json	JSON Support
..and many more	

Differnece between Spring Framework and Spring Boot

Diagram



==========

session 19

==========

=> Spring boot is a spring based project that provides AutoConfigurations

(Reduce common code writeen in Spring f/w)

=> Spring boot has provided starters (pre-defined JARs for every concept)

ex: spring-boot-starter-jdbc

spring-boot-starter-data-jpa

.... etc

=> Spring Boot applications must connect with parent Project our project is known as child project.

=> If we use maven project there will be pom.xml

for Gradle build.gradle files exist.

=> Spring boot do not support XML configuration.

we need to work on java and Annotation Configuration.

xml is slow compared to java/annoation.

=> Spring boot has provided Embeeded Database (H2, HyperSQL, Derby)

These are also called as In-Memory Database(No Download + No Install)

Recomanded to use dev/test Environment. Not in Production.

=> Spirng boot has provided Embedded Webservers:

(Apache Tomcat(default), Eclipse Jetty, JBoss Undertow)

to run our application (No downlead + No Install)

---

https://start.spring.io/

Spring Initializer

=> This website provides support to create spring boot application.

-> Even if we use Any IDE (Eclipse, STS, VScode, IntellijIDea) that should be finally connected to spring Initializer.

==========

session 20

=======

=======

Spring Boot - Project Structure and basics.

==========

step to create project:

> File> new > Spring Starter project> Enter Details

Type of files : There are mainly 3 types of Files in Every Spring Boot application, gives as:

1. Starter class /Main Class

=> This is an entry class in execution.

It create spring Container (new Container : ApplicationContext(I)

Impl: AnnotationConfigApplicationContext(C)

=> It load all classes using basePackage All the classes which are in basePackage or subPackage are scanned by Spring Container and create objects.

=> provide data, read properties file, inject data, link object..etc

---

2. Input File (application.properties/application.yml)

=> This file is used to pass input to container objects.

=>It holds data in Key=value format

=>keys are pre-defined here, we can pass our own key(custom keys).

=>Internally every key is a variable

=> we can find all pre-defined key list at:

https://docs.spring.io/spring-boot/appendix/application-properties/index.html#appendix.application-properties.cache

=> names of these files are fixed.

But we can even provide our custom properties files.

=>A key name can have symbols like dot(.) dash(-) and underscore(_) No other symbols allowed

ex : spring.datasource.driver-class-name=

my.app-name_model=

---

3. pom.xml

POM Stand for Porject Object Model.

=> The pom.xml file is the configuration file or a Maven project. It containers project information, dependencies, plugins, properties, and build configuration required to build and manage the application.

Advangtages of pom.xml

Easy dependency management

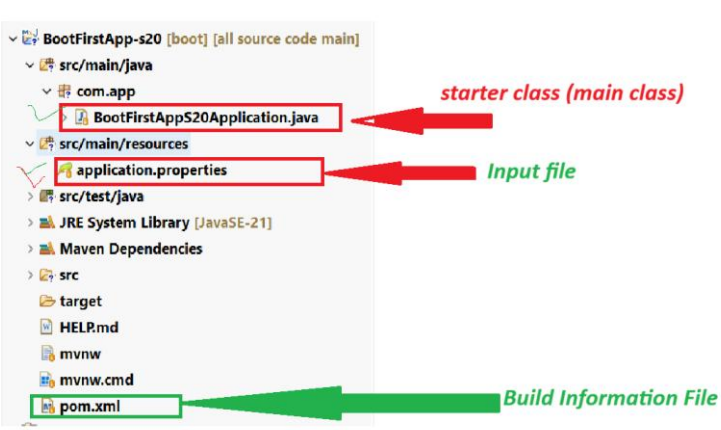
Automatic Library download

Easy project bild process.

Better version management

Reduce manual configuration.

Diagram



---

session 21

==========

=========

Runners:

=>Runners are used to execute any logic only one time, while starting the application (i.e after creating the spring container);

example:

Setup Data in Database.

insert RolesTable(ADMIN, USER, GUEST)..etc

- Load File and read data..etc

-Test some logic..etc

How Runners work in Spring Boot?

=> we need to define one class that should implement interface CommandLineRunner.

=> we need to override abstract method run() and define our logic.

=> Main class/starter class, after creating container, executing all runners classes.

=> we can define multiple runners, and we can provide our own execution order using @Order(value) annotation.

=> @Order(_) lowest value is executed first. if we did not specify any Order then executed at last.

==========

ex Code

project Name : BootRunnerEx-s21

MessageRunner.java

```
package com.app.runner;


import org.springframework.boot.CommandLineRunner;
import org.springframework.core.annotation.Order;
import org.springframework.stereotype.Component;
@Component
@Order(5)
public class MessageRunner implements CommandLineRunner{
```

```
@Override
public void run(String... args) throws Exception {
System.out.println("FROM MESSAGE RUNNER");

}

}
```

EmailRunner.java

```
package com.app.runner;

import org.springframework.boot.CommandLineRunner;
import org.springframework.core.annotation.Order;
import org.springframework.stereotype.Component;

@Component
@Order(2)
public class EmailRunner implements CommandLineRunner {

@Override
public void run(String... args) throws Exception {
System.out.println("FROM EMAIL RUNNER");

}

}
```

JdbcRunner.java

```
package com.app.runner;

import org.springframework.boot.CommandLineRunner;
```

```
import org.springframework.core.annotation.Order;

import org.springframework.stereotype.Component;


@Component

@Order(3)

public class JdbcRunner implements CommandLineRunner  {


@Override

public void run(String... args) throws Exception {

System.out.println("FROM JDBC RUNNER");


}


}
```

---

output

FROM EMAIL RUNNER

FROM JDBC RUNNER

FROM MESSAGE RUNNER

==========

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==========

==========

@Value : This annotatio is used to read data from Properties file into variable.

If we have n-variable in class then we need to provide n-@Value annotation.

@ConfigurationProperites:

=> It is used to load multiple key=value data into variable

=> Must provide one common prefix for key name.

=> Same prefix mubst be matched while loading keys in class

=> key name and variable name must be matching.

=> Must generate setter/getter methods.

=> If we don't follow above rules then spring boot will not load key=val data into variables. No Error, No Exception. variable holds default values.

example project

BootConfigPropsEx-s22

1. Spring Bean

```
package com.app.service;

import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

import lombok.Data;

@Data
@Component
@ConfigurationProperties(prefix = "my.app")
public class EmailService {

private String host;
private int port;
private String username;
private boolean active;

}
```

---

2. application.properties

```
spring.application.name=BootConfigPropsEx-s22

my.app.host=GmailServer
my.app.port=586
my.app.username=IndiaFreeInternship
my.app.active=false
```

---

3. Runner class

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

import com.app.service.EmailService;

@Component
public class TestObjRunner implements CommandLineRunner {

    @Autowired
private EmailService emailService;

@Override
public void run(String... args) throws Exception {
System.out.println(emailService);

}

}
```

---

output

```
EmailService(host=GmailServer, port=586, username=IndiaFreeInternship, active=false)
```

@ConfigurationProperties with Collections

=>It Support loading Collections data from properties file of type List,Set,Array,Map and Properties

=> for List/Set/Array

syntax is

prefix.variable[index]=value

=> For map/Properties syntax is

prefix.variable.mapKey=mapValue

=> @ConfigurationProperties even support loading complex reference type data using prefix.refVariable.variable=value syntax.

Example project BootConfigPropsEx-s22

==========

1. Spring bean

```
package com.app.service;

import java.util.Map;
import java.util.Properties;


import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

import lombok.Data;

@Data
@Component
@ConfigurationProperties(prefix = "my.app")
public class EmailService {

private String host;
private int port;
private String username;
private boolean active;
```

```
// private List<String> models;

//private Set<String> models;

private String[] models;

//private Map<String,String> data;

private Properties data;


private Message mob;


}
```

---

2. Spring bean

```
package com.app.service;


import lombok.Data;


@Data

public class Message {


private String code;

private String details;


}
```

---

application.properties

```
spring.application.name=BootConfigPropsEx-s22


my.app.host=GmailServer

my.app.port=586

my.app.username=IndiaFreeInternship

my.app.active=false
```

```
my.app.models[0]=M1
my.app.models[1]=M2
my.app.models[2]=M3
my.app.models[3]=M2

my.app.data.C1=ABC
my.app.data.C2=XYZ
my.app.data.C3=pqr

my.app.mob.code=C1325
my.app.mob.details=SampleMobile
----------
```

TestObjRunner.java

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

import com.app.service.EmailService;

@Component
public class TestObjRunner implements CommandLineRunner {

    @Autowired
private EmailService emailService;

@Override
public void run(String... args) throws Exception {
System.out.println(emailService);

}
```

}

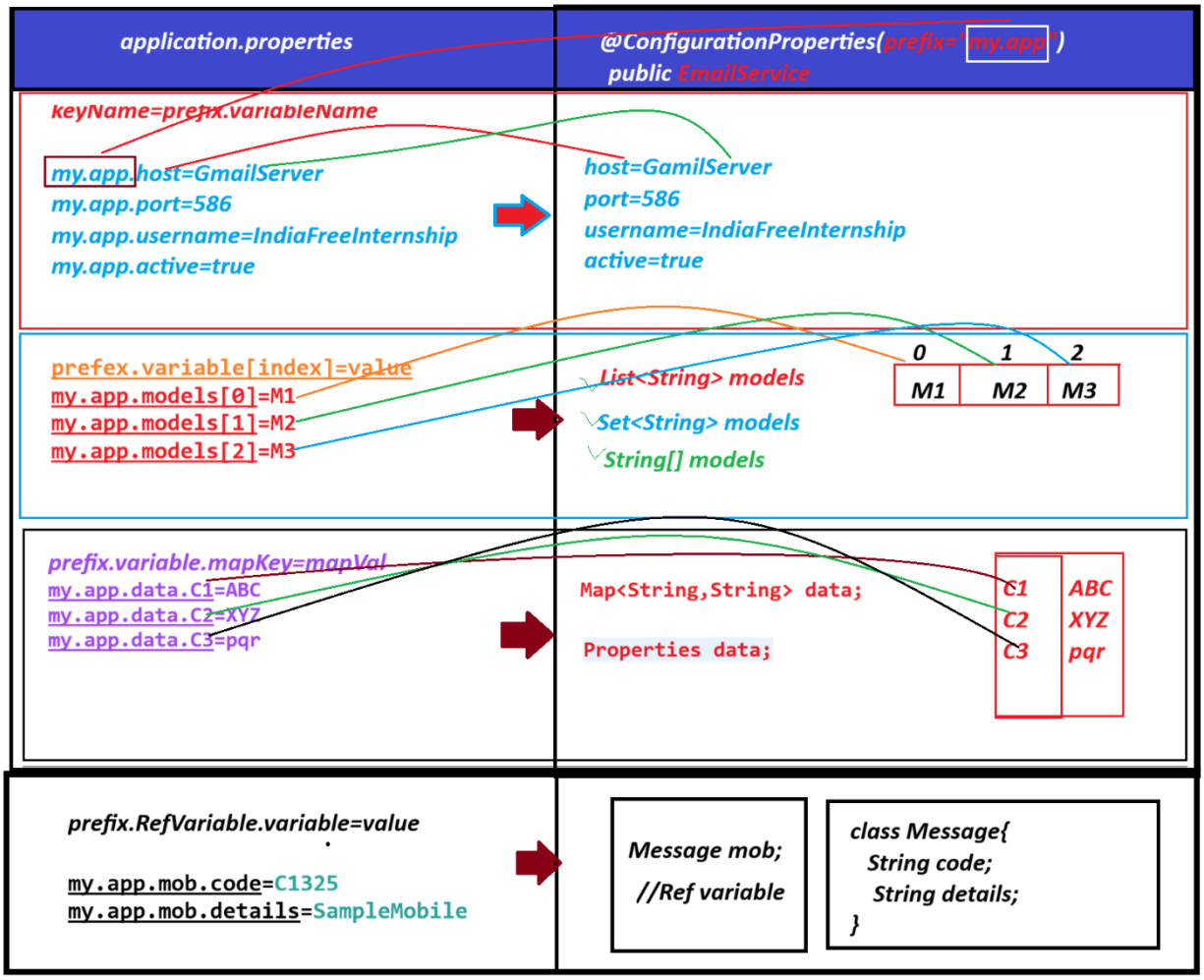
---

output

EmailService(host=GmailServer, port=586, username=IndiaFreeInternship, active=false, models=[M1, M2, M3, M2], data={C3=pqr, C1=ABC, C2=XYZ}, mob=Message(code=C1325, details=SampleMobile))

==========

Diagram



======

==========

YAML (Yes Another Markup Language)

YAMALiant Language

=>YAML is a new way of represening KEY=VAL paris.

=> In spring boot we can load data using Properties file and YAML file

=> YAML file extendsion is .yml

=> In YAML file we do not write duplicate prefix/levels.

=> It is more readable, faster in parsing, supports single file profiels.

=> snake YAML(Jar name)

==========

example project BootConfigYAMLEx-s22

1. Spring bean

```
package com.app.service;


import java.util.Map;
import java.util.Properties;



import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;


import lombok.Data;


@Data
@Component
@ConfigurationProperties(prefix = "my.app")
public class EmailService {


private String host;
private int port;
private String username;
private boolean active;


// private List<String> models;
//private Set<String> models;
//private String[] models;
//private Map<String,String> data;
```

```
//private Properties data;


//private Message mob;


}
```

---

2. Spring bean

```
package com.app.service;


import lombok.Data;


@Data
public class Message {


private String code;
private String details;


}
```

---

application.yml

```
my:
 app:
  host: GmailServer
  port: 586
  username: IndiaFreeInternship
  active: true
```

TestObjRunner.java

```
package com.app.runner;
```

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

import com.app.service.EmailService;

@Component
public class TestObjRunner implements CommandLineRunner {

    @Autowired
private EmailService emailService;

@Override
public void run(String... args) throws Exception {
System.out.println(emailService);

}

}
```

---

output

```
EmailService(host=GmailServer, port=586, username=IndiaFreeInternship, active=true)
==========
==========
```

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YAML (Yet Another Markup Language)

YAMALiant Language

=>YAML is a new way of represening KEY=VAL paris.

=> In spring boot we can load data using Properties file and YAML file

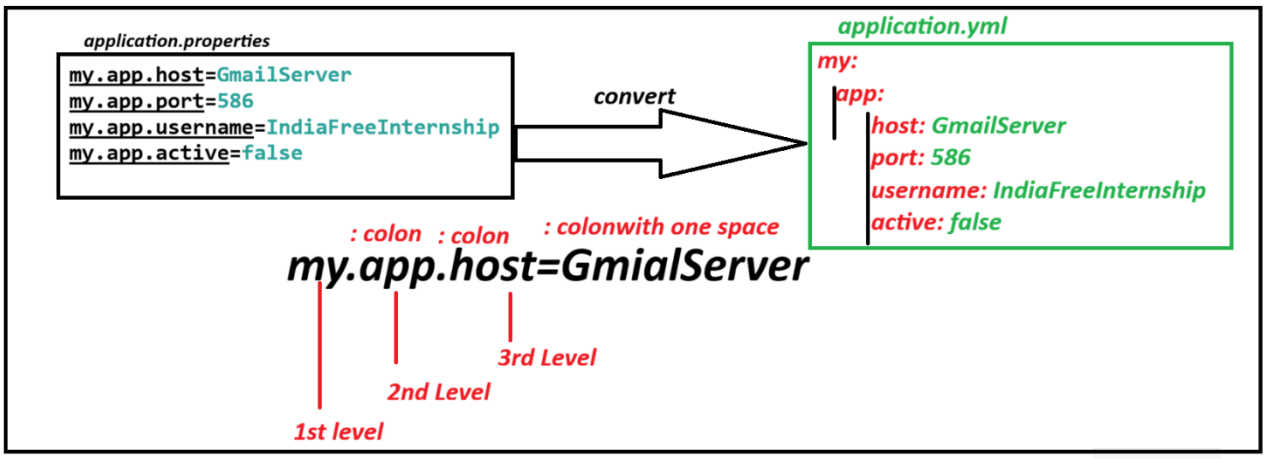
=> YAML file extendsion is .yml

=> In YAML file we do not write duplicate prefix/levels.

=> It is more readable, faster in parsing, supports single file profiles.

=> snake YAML(Jar name)

Diagram



Notes:

1. Use symbols colon (:) in the place of dot(.) and equals(=)

2. After every colon (new level) change line for next level.

3. we need to provide spaces for everynew level (not for 1st level)

4. Space count must match propoer indentation for the given level.

5. Must provide colon and space before value:

=> if we do not follow space rules or any other then spring boot throws (Snake YAML Exception) : ScannerException.

application.properties

---

my.app.driver=Oracle

my.app.url=JDBC-ORCL

my.app.model=NEW

---

application.yml

my:

  app:

driver: Oracle

url: JDBC-ORCL

model: NEW

==========

For List/Set/Array Format: (here dash - indicates index number)

key:

- value1

- value2

- value3

==========

For Map/ Properties /ref Type

mapKey: mapValue

==========

Example program:

BootConfigYAMLEx-2-s23

---

1 Spring bean

ProductInfo.java

```
package com.app.bean;

import java.util.List;
import java.util.Map;

import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

import lombok.Data;

@Data
@Component
```

```
@ConfigurationProperties(prefix = "my.app")
public class ProductInfo {

private String pcode;
private int pid;
private double pcost;

private List<String> colors;
private Map<String, String> models;
private Vendor vob;
}
```

---

2. Spring Bean

```
package com.app.bean;

import lombok.Data;

@Data
public class Vendor {

private int vid;
private String vcode;
}
```

---

3. Runner class

TestProductRunner.java

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;
```

```
import com.app.bean.ProductInfo;


@Component
public class TestProductRunner implements CommandLineRunner {
   @Autowired
private ProductInfo productInfo;


@Override
public void run(String... args) throws Exception {
System.out.println(productInfo);


}
}
```

---

application.yml

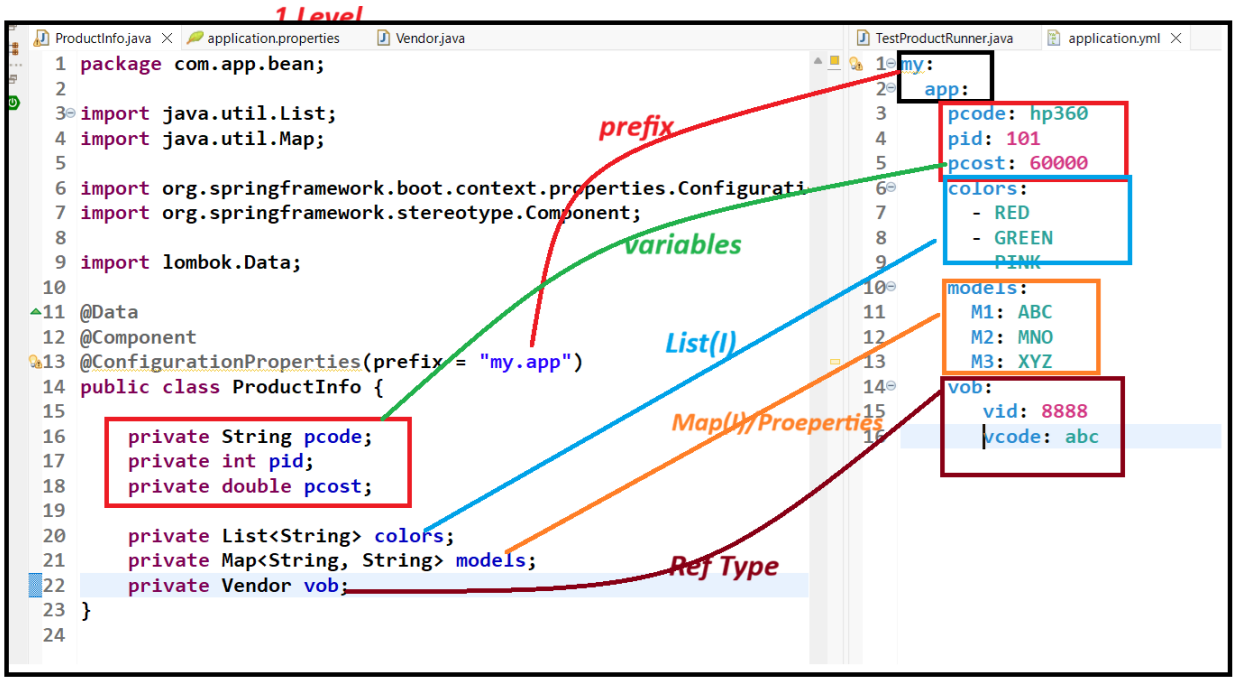
```
my:
  app:
    pcode: hp360
    pid: 101
    pcost: 60000
    colors:
      - RED
      - GREEN
      - PINK
    models:
      M1: ABC
      M2: MNO
      M3: XYZ
    vob:
       vid: 8888
       vcode: abc
```

==========

output

ProductInfo(pcode=hp360, pid=101, pcost=60000.0, colors=[RED, GREEN, PINK], models={M1=ABC, M2=MNO, M3=XYZ}, vob=Vendor(vid=8888, vcode=abc))

Diagram



==========

Example 1

application.properties

----

my.app.test=A

spring.boot.version=4.1.0

my.app.model=ONE

spring.boot.format=MAVEN

----

application.yml

my:

app:

test: A

model: ONE

spring:

boot:

version: 4.1.0

format: MAVEN

Example 2

application.properties

my.app.version=4.0

my.test.model=New

my.app.format=OK

my.test.info=WELCOME

----

application.yml

my:

app:

version: 4.0

format: OK

test:

model: New

infor: WELCOME

==========================

Online converter

https://www.xkit.io/devtools/properties-to-yaml

==========================

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=========

=========

YAML : complex Tyes(List/Map/Ref Type)

List/Set/Array :

prefix:

variable:

- value1 # dash<space>value

- value2

- value3

Map/Properties

prefix:

variable:

mapKey:

mapValue # key:<space>value

mapKey:

mapValue

---

prefix:

hasAVariable:

variable: value

variable: value

==========

Q) can we define both properties and yaml file in springboot application?

Ans> YES, We can define both application.properties and applicaton.yml

Q) what is both file having same key name and different values?

Ans> Always spring Container takes value from application.properties with high prirority.

example program

name : BootConfigYAMLEx-s24

dependency : Lombok

1. Spring bean

package com.app.bean;

import lombok.Data;

@Data

public class Publisher {

```
                                                                    private Integer
pid;
                                                                    private String
code;
                                                                    private boolean
active;
}
----------
package com.app.bean;

import java.util.Map;
import java.util.Set;

import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

import lombok.Data;

@Data
@Component
@ConfigurationProperties(prefix = "my.book")
public class BookInfo {

                                                                    private String
bname;
                                                                    private int
bcost;

                                                                    //private
List<String> authors;
                                                                    //private
Set<String> authors;
                                                                    private String[]
authors;
```

```
                                                                    private
Map<String,String> versions;

                                                                    private
Publisher pob; // HAS-A Variable



}
```

---

2. application.yml

```
my:
  book:
    bname: science
    authors:
     - HC VERMA
     - XYZ
     - PQR
     - HC VERMA
    versions:
     v1: E1
     v2: E2
     v3: E3
    pob:
     pid: 90012
     code: PC12345
     active: true
```

---

3. Runner class

BookRunner.java

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
```

```
import org.springframework.stereotype.Component;

import com.app.bean.BookInfo;

@Component
public class BookRunner implements CommandLineRunner{

                                                    @Autowired
                                                    private
BookInfo bookInfo;

                                                    @Override
                                                    public void
run(String... args) throws Exception {

                                                    System.out.prin
tln(bookInfo);

                                                    //System.out.pr
intln(bookInfo.getAuthors().getClass().getName());
                                                    }

}
==================
output
BookInfo(bname=science, bcost=400, authors=[HC VERMA, XYZ, PQR, HC VERMA], versions={v1=E1,
v2=E2, v3=E3}, pob=Publisher(pid=90012, code=PC12345, active=true))
--------------------------------------------------------------------------------
 class Author{
                                                    String name;
                                                    String addr;
 }

 class Book{
```

List<Author>

authors; // multiple authors objects

Map<String,Authro> authors: /multiple author objects

}

YAML File input must be given.

Environments:-

An environment is a separate setup or configuration where an application is developed, tested, or deployed for a specific purpose.

1. Dev Environment: Developer machine

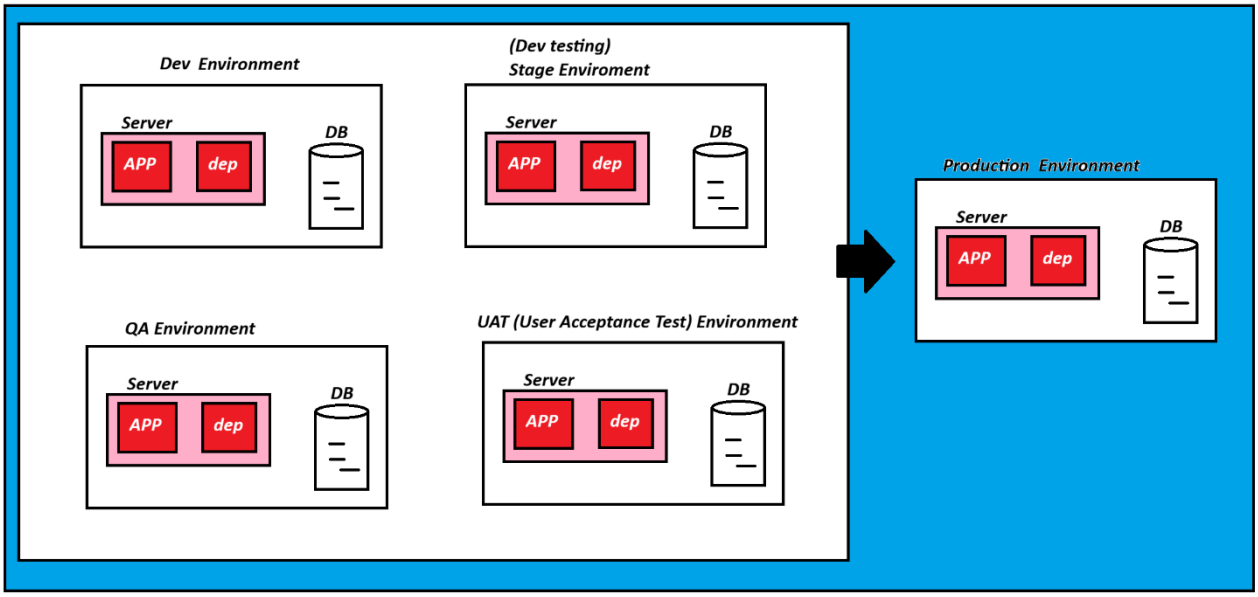
2. QA Team : Testing Team

3. State Env : Developer Test

4. UAT Env : User Acceptance Test (Client Testing)

5. Production Env : Production Env (Actual server given to EndCustomer)

Diagram



---

profiles

Maintain multiple Properties/YAML files for multiple environments.

Based on Environment selected/running, matching properties/yaml file will be loaded

Rule Name : application-{profilename}.properties

Activate : (CMD) --spring.profiles.active={profilename}

examples:

Dev mahcine (default profile) ===> applicaiton.properties or application.yml

UAT machine ===> application-uat.properties or application-uat.yml

Production machine ===> application-prod.properties or application-prod.yml

our application code remains same in almost all environments.

But properties may get varied. Ex: username and password or database.

==========

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==========

Spring Boot Profiles

Maintain multiple Properties/YAML files for multiple environments.

based on Environment selected/running, matching properties/yaml file will be loaded.

Name Rule : application-{profileName}.properties

Activate : (CMD) --spring.profiles.active={profileName}]

How It works

(Steps)

1. Developer writes code (same code for all env)

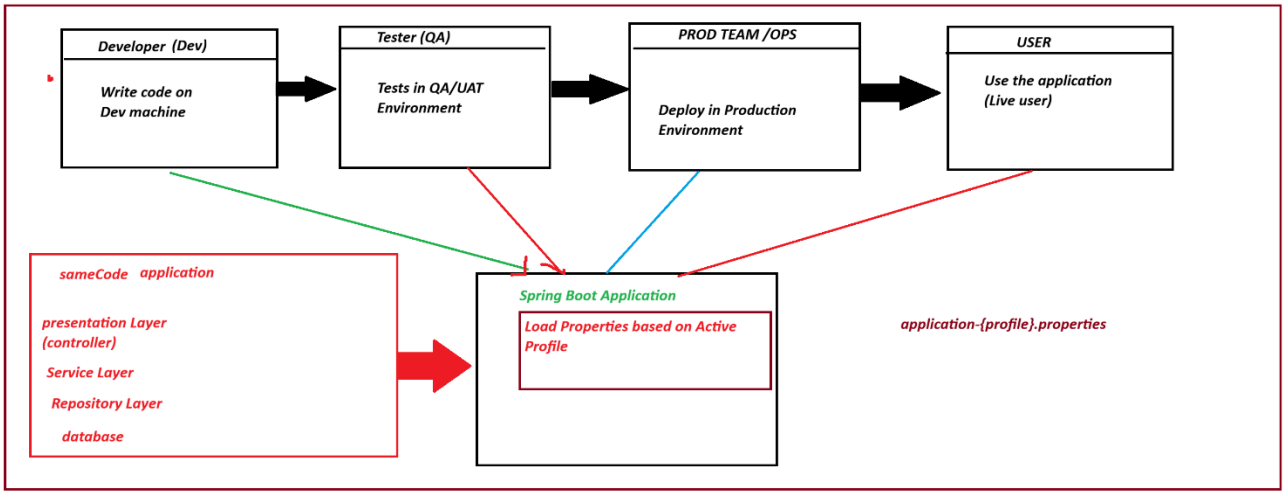
2. Create multiple property files for different environments

3. Run application with Active profile using command line.

4. spring boot loads default file + matching profile file.

5. If any property missing in active profile -> take from default file.

6. Application connects to respective database.



example program

BootProfleEx-s25

Dependency Lombok

1. Spring bean

```
package com.app.bean;

import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

import lombok.Data;
@Data
@Component
@ConfigurationProperties(prefix = "my.app.db")
public class MyDbCon {

private String driver;
private String url;
private String username;
private String password;
private String database;
}
```

---

2. Input files

application.properties

```
spring.application.name=BootProfleEx-s25
my.app.db.driver=Oracle
my.app.db.url=JDBC-ORCL
my.app.db.username=system
my.app.db.password=root
my.app.db.database=testdb
```

---

application-qa.properties

```
my.app.db.driver=MySQL
my.app.db.url=JDBC-MySQL
my.app.db.username=root
my.app.db.password=root
```

---

application-prod.properties

```
my.app.db.driver=Oracle
my.app.db.url=JDBC-ORCL-PROD
my.app.db.username=TEST
my.app.db.password=sample
my.app.db.database=testdb
```

==========

3. Runner Class

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

import com.app.bean.MyDbCon;
```

```
@Component
public class TestRunner implements CommandLineRunner {
@Autowired
private MyDbCon myDbCon;

@Override
public void run(String... args) throws Exception {
System.out.println(myDbCon);

}

}
```

---

output

```
The following 1 profile is active: "prod"
MyDbCon(driver=Oracle, url=JDBC-ORCL-PROD, username=TEST, password=sample, database=testdb)
```

==========

Right click main class code >Run As > Run Configuration.... > click Argument Tab> go to > program Arguments

Enter data like

--spring.profiles.active=qa

apply and Run

==========

Archive : Grouping all files as a single unit.

JAR (Java Archive)=> use for Standalone application

file1.java

file2.java

file3.java

file4.java

Archive

==========

WAR=> Web Archive

.class+html+css+javaScript+Images==> .war (web application)

==========

developer>.java>Compile>.class>.jar(build)

==========

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==========

Profiles using YAML Files

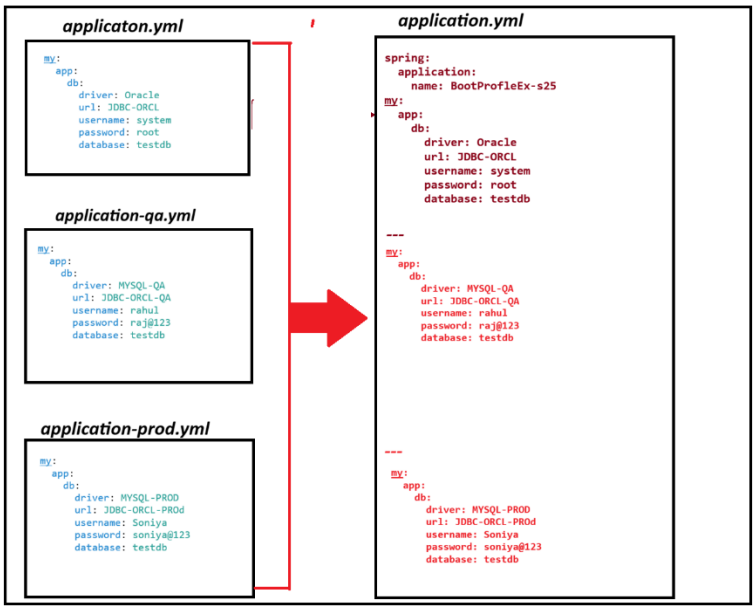
=> For properties file, we need define multiple files in case of Profiles concept. n-profiles=n-properties files;

YAML files too n-profile=n-YAML file

=> YAML even support single file for Multiple Profiles concept, using 3 dash sysmobs(---).

=> For every area we need to specify profile name (or) on which profile it should be loaded, using key.

```
spring:
  config:
    active:
on-profiles:
      - profile1
  - profile2
  - profile3
```



==========

Example project : BootProfleYAMLEx-s26

Dependency Lombok

1. Spring bean

```
package com.app.bean;


import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;


import lombok.Data;
@Data
@Component
@ConfigurationProperties(prefix = "my.app.db")
public class MyDbCon {


private String driver;
private String url;
private String username;
private String password;
private String database;
}
```

---

2. Input files

application.yml

```
spring:
 application:
   name: BootProfleEx-s25
my:
 app:
   db:
     driver: Oracle
     url: JDBC-ORCL
```

```
    username: system
    password: root
    database: testdb


---
 my:
  app:
   db:
     driver: MYSQL-QA
     url: JDBC-ORCL-QA
     username: rahul
     password: raj@123
     database: testdb
 spring:
   config:
    active:
     on-profile: qa
---
my:
 app:
  db:
   driver: MYSQL-Prod
   url: JDBC-ORCL-prod
   username: soniya
   password: soniya@9999
   database: testDb
spring:
 config:
  active:
    on-profile: prod
=====================
```

3. Runner Class

```
package com.app.runner;


import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.boot.CommandLineRunner;

import org.springframework.stereotype.Component;


import com.app.bean.MyDbCon;


@Component

public class TestRunner implements CommandLineRunner {

@Autowired

private MyDbCon myDbCon;


@Override

public void run(String... args) throws Exception {

System.out.println(myDbCon);


}


}
```

==========

output

MyDbCon(driver=MYSQL-Prod, url=JDBC-ORCL-prod, username=soniya, password=soniya@9999, database=testDb)

==========

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======

@Autowired usecases

=> If Two Spring Beans are connected using Reference Type, (Two classes are connected using HAS-A relation)

then @Autowired annotation create link between them.

=> If spring container found one child object, then for @Autowired two object are linked.

=> If Spring container found zero child objects to link for Autowired then container throws exception:

UnsatisfiedDependencyException:

---

org.springframework.beans.factory.UnsatisfiedDependencyException: Error creating bean with name 'product': Unsatisfied dependency expressed through field 'vod': No qualifying bean of type 'com.app.bean.Vendor' available: expected at least 1 bean which qualifies as autowire candidate. Dependency annotations: {@org.springframework.beans.factory.annotation.Autowired(required=true)}

=> If we define @Autowired(required=false) then it will inform container if zero mathcings found. Please keept it as null value.

==========

examlple project BootaAutowiredEx-s27

dependency Lombok

---

1. spring bean

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
import lombok.ToString;

@Component
@ToString
public class Product {

@Autowired(required = false)
//@Autowired(required = true)
private Vendor vod;   // HAS-A

}
```

---

2. spring bean

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

import lombok.ToString;

@Component
@ToString
public class Vendor {

@Value("v1-xyz")
private String vname;

}
```

---

3. Runner class

```
package com.app.runner;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

import com.app.bean.Product;

@Component
public class TestRunner implements CommandLineRunner {

@Autowired
private Product pob;
```

```
@Override
public void run(String... args) throws Exception {
    System.out.println(pob);
}

}
```

---

java Config : Two define multiple objects create obj for a pre-defined class

1. write one public class
2. Apply @Configuration (Input to Container)
3. Define one method for one object

```
public <ClassName> <objName>(){
}
```

==========code==========

1. spring bean

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Qualifier;
import org.springframework.stereotype.Component;

import lombok.ToString;

@Component
@ToString
public class Product {

//@Autowired(required = false)
//@Autowired(required = true)
```

```
@Autowired
//@Qualifier("vod2")
private Vendor vod;   // HAS-A

}
```

2.config

```
package com.app.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.context.annotation.Primary;

import com.app.bean.Vendor;

@Configuration
public class MyAppConfig {

@Bean
@Primary
public Vendor vod1() {
Vendor v1=new Vendor();
v1.setVname("v1-abc");
return v1;

}

@Bean
public Vendor vod2() {
Vendor v1=new Vendor();
```

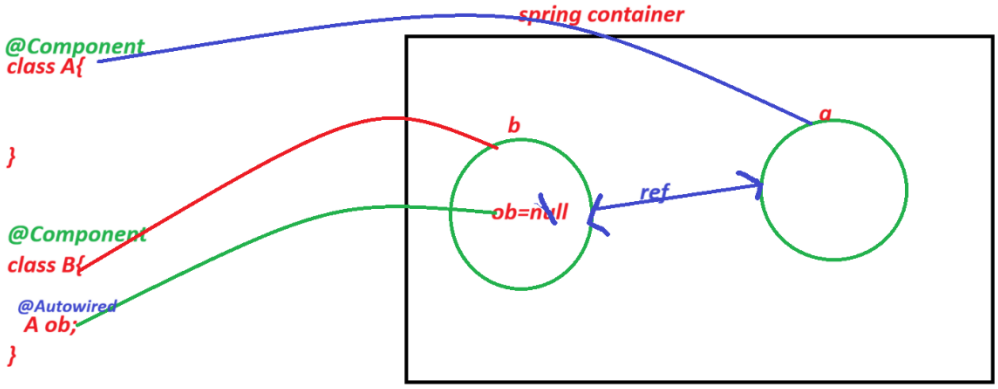
```
v1.setVname("v1-pqr");
return v1;
}
}
-------
```

spring bean

```
package com.app.bean;

import org.springframework.beans.factory.annotation.Value;
import lombok.Data;
@Data
public class Vendor {
private String vname;
}
```

Diagram



0 objects found	@Autowired -> @Autowired(required=true) NoSuchBeanDefinitionException @Autowired(required=false)--> keep it as null
1 obj found	If one child object is found, container will link them (No Error, No Exception, we get output)
1 obj found	If two or more child objects found then NoUniqueBeanDefinitionExcetpion

solution:

1. byName:
   we can modify one child bean name as Ref variable name
2. manual Select ****
   use @Qualifier("objname") will search and link that child obj
3. Default obj | Recommended obj
   @Primary says if multiple child beans found then select this by default

---

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==========